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Measurement of inclusive $B \to \Lambda_c$ branching fractions using Belle data and hadronic Full Event Interpretation

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Abstract

Inclusive $B \to \Lambda_c$ branching fractions were measured most recently by BaBar collaboration. However, the measurement still presented a poor accuracy. A more precise measurement of inclusive $B \to \Lambda_c$ branching fraction could be useful to gain a better confidence on B meson weak decays treatment. With help of the Full Event Interpretation algorith, it is possible to perform a more precise measurment of inclusive $B \to \Lambda_c$ branching fractions using Belle data set.

Contents

Chapter 1

Introduction

Inclusive B meson baryonic decays with a Λ_c baryon in the final state are the most abundant, due to a relatively large V_{cb} element of the CKM matrix. The *BaBar* experiment measured their branching fractions to be around the percent level (see ref. [1]). However, the branching fractions were determined with big uncertainties: nearly 50% on the measured values or, in the case of the $B^0 \to \Lambda_c^+$ decay, only an upper limit could be established. A more precise measurement of inclusive $B \to \Lambda_c$ branching fractions may shed light on the appropriateness of B meson weak decays treatment, particularly of strong interaction effects modelling. Predictions for inclusive branching fractions are given, for example, in ref. [2] or in [3] for $B \to \Lambda_c p$ decays.

Exploiting the Full Evenet Interpretation (FEI) algorithm, developed for the Belle II experiment, it may be possible to perform a more precise measurement of inclusive $B \to \Lambda_c$ branching fractions, using the full Belle data set. A more precise measurement may also trigger further research on currently scarce theory predictions for B meson decays to charm baryons.

1.1 Analysis Setup

The reconstruction is performed with `BASF2` release `05-02-03` together with the `b2bii` package in order to convert the *Belle* `MDST` files (`BASF` data format) to *Belle II* `MDST` files (`BASF2` data format). The FEI version used is `FEI_B2BII_light-2012-minos`.

1.2 Datasets

The Belle detector acquired a dataset of about $L_0 \approx 710 fb^{-1}$ of integrated luminosity in its lifetime at the $\Upsilon(4S)$ energy of 10.58 GeV, which corresponds to about 771 $\times 10^6 B\bar{B}$ meson pairs. Additionally, several streams of Monte-Carlo (MC) samples were produced, where each stream of MC corresponds to the same amount of data that was taken with the detector. No specific signal MC was used: instead of producing dedicated signal MC samples, the samples were obtained by filtering the decays of interest from the generic on-resonance MC samples. The following samples were used in this analysis:

- data
- MC - 6 streams of B^+B^- and $B^0\bar{B}^0$ (denoted as `charged` and `mixed`) for signal decays and backgrounds (if more of the in total existing 10 streams is used it is explicitly specified throughtout this note).
  - 6 streams of $q\bar{q}$ produced at $\Upsilon(4S)$ resonance energy
  - 6 streams of $q\bar{q}$ produced at 60 MeV below $\Upsilon(4S)$ resonance energy, where each stream corresponds to $1/10 \times L_0$.

Chapter 2

Event selection and reconstruction

In this chapter the procedure for reconstruction of the events where one B meson decays inclusively to a Λ_c baryon and the accompanying B meson decays hadronically is illustrated.

2.1 B_{tag} reconstruction

The FEI is an exclusive tagging algorithm that uses machine learning to reconstruct B meson decay chains and calculates the probability that these decay chains correctly describe the true process. In this analysis only hadronically reconstructed decay chains are considered. The training called `FEI_B2BII_light-2012-minos` is used. Tag-side B meson candidates are required to have a beam-constrained mass greater than 5.22 GeV/c^2 and $-0.15 < \Delta E < 0.07$ GeV.
In the case of multiple candidates in the same event, the candidate with the highest SignalProbability (the signal probability calculated by FEI using FastBDT) is chosen. To suppress the background constisting of B^0 events misreconstructed as B^+ (and vice-versa) from neutral (charged) decays also a B^0 (B^+) candidate is reconstructed with FEI and if its SignalProbability is higher than the charged (neutral) reconstructed B meson, the event is discarded. This constitutes a sort of crossfeed-veto, rejecting part of events belonging to the other typology of decays of interest: for example in the case one is interested in reconstructing $B^{+/-}$ decays and the event actually contains $B^0/\bar{B}^0$ decays, the FEI reconstructed neutral B meson candidate most likely presents a higher SignalProbability than the charged FEI reconstructed candidate.

2.2 Λ_c reconstruction

In the *rest of event* (ROE) of the reconstructed B_{tag} meson, to select $\Lambda_c \to pK\pi$ signal candidates, the following event selection criteria are applied (same PID cuts were used for example in the Belle Note 1521 `https://belle.kek.jp/secured/belle_note/gn1521/BN_v1.pdf`). Charged tracks with the impact parameters perpendicular to and along the nominal interaction point (IP) are required to be less than 2 cm and 4 cm respectively

($dr < 2$ cm and $|dz| < 4$ cm).
The pion tracks are required to be identified with $\frac{\mathcal{L}_\pi}{\mathcal{L}_K+\mathcal{L}_\pi} > 0.6$. The kaon tracks are required to be identified with $\frac{\mathcal{L}_K}{\mathcal{L}_K+\mathcal{L}_\pi} > 0.6$, and the proton/anti-proton tracks are required to be identified with $\frac{\mathcal{L}_{p/\bar{p}}}{\mathcal{L}_K+\mathcal{L}_{p/\bar{p}}} > 0.6$ and $\frac{\mathcal{L}_{p/\bar{p}}}{\mathcal{L}_\pi+\mathcal{L}_{p/\bar{p}}} > 0.6$, where the $\mathcal{L}_{\pi,K,p/\bar{p}}$ are the likelihoods for pion, kaon, proton/anti-proton, respectively, determined using the ratio of the energy deposit in the ECL to the momentum measured in the SVD and CDC, the shower shape in the ECL, the matching between the position of charged track trajectory and the cluster position in the ECL, the hit information from the ACC and the dE/dx information in the CDC.
For the Λ_c candidates a vertex fit is performed with `TreeFitter`, requiring it to converge. If there are more than one Λ_c combination, then the best candidate based on the χ^2 probability is chosen. The Λ_c signal region is defined to be $|M_{\Lambda_c} - m_{\Lambda_c}| < 20$ MeV/c^2 ($\sim 3\sigma$), here m_{Λ_c} is the nominal mass of m_{Λ_c}.

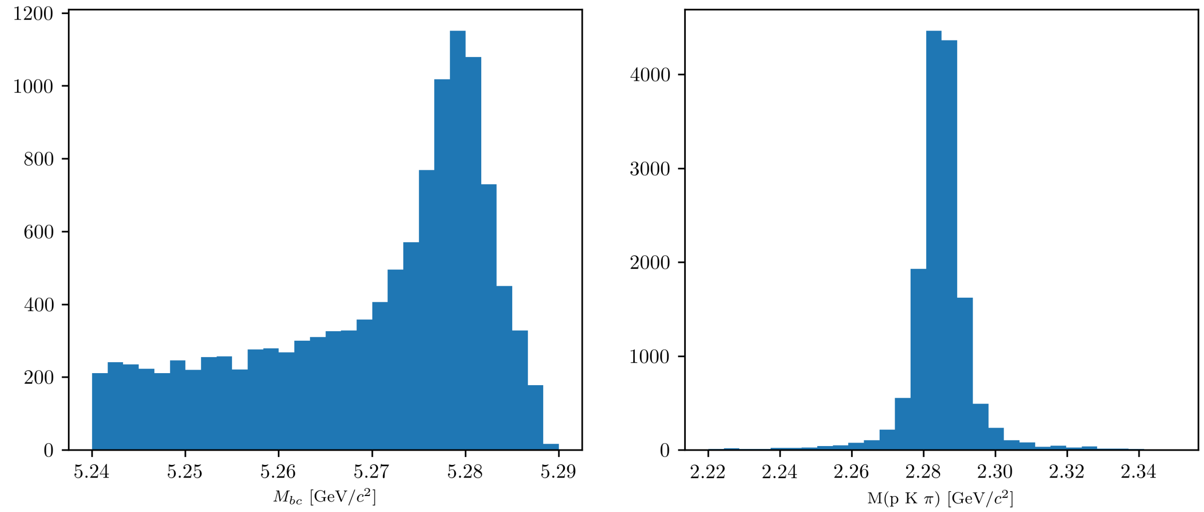


Figure (2.1) M_{bc} and $M(pK\pi)$ distributions of B_{tag} and Λ_c candidates reconstructed in the signal sample.

2.3 Wrongly reconstructed B_{tag} candidates

In the case of the signal sample the distributions for the beam-constrained mass M_{bc} and for the correctly reconstructed Λ_c candidates, look like in Fig. 2.1. If one then investigates the M_{bc} distribution of the B_{tag} candidates reconstructed with FEI, it can be seen that there is a peaking structure for wrongly reconstructed B mesons (as in Fig. 2.2), according to the BASF2 internal truth matching variable **isSignal**. It is obvious from this that the BASF2 internal truth matching variable cannot be used to separate properly the signal events in correctly and wrongly reconstructed B mesons. In the study BELLE2-NOTE-TE-

2021-026 `https://docs.belle2.org/record/2711/files/BELLE2-NOTE-TE-2021-026.pdf` a possible solution was found developing new variables that can be used for an improved truth matching for the FEI (those variables were added to a newer BASF2 release than the one used for this study). In the present study instead a more "traditional" approach was adopted: fitting the M_{bc} distribution with a sum of PDFs that account for the flat (background) component and the peaking (signal) component. The first component represents the combinatorial background, i.e. B mesons that were mis-reconstructed, and therefore those events are denoted from now on as "**misreconstructed signal**". The peaking component represents the correctly reconstructed signal events in M_{bc} and therefore denoted from now on as "**reconstructed signal**". Only the second one is then considered for the signal yield, while the first is counted as a background. To validate this method a control decay study was performed on the flavor correlated $B^+ \to \bar{D}^0$ channel.

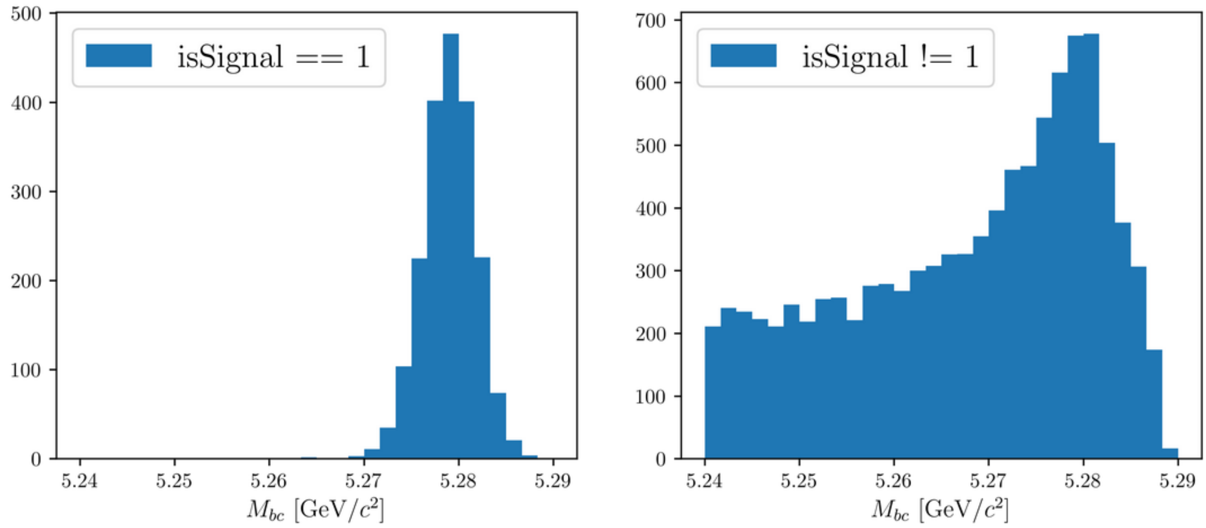


Figure (2.2) M_{bc} distribution of B_{tag} candidates reconstructed in the signal sample, truth-matched (on the left) and not (on the right).

2.4 Signal selection optimization

To further enhance the purity of the signal decays, an optimization procedure is adopted to determine optimal cuts for a set of variables for each decay mode under investigation by this study. The cuts on the following variables are optimized:

- $foxWolframR2$: the event based ratio of the 2-nd to the 0-th order Fox-Wolfram moments

- SignalProbability: the already mentioned signal probability calculated by FEI using FastBDT

- $p_{CMS}^{\Lambda_c}$: momentum of the Λ_c candidates in the center of mass system

The optimization is based on the Figure Of Merit (FOM): FOM $= \frac{S}{\sqrt{S+B}}$

Where S and B are respectively signal and background events in the signal region: $M_{bc} > 5.27$ GeV$/c^2$, $2.2665 < M(pK\pi) < 2.3065$ GeV$/c^2$.
Due to the issue reported in Sec. 2.3, to separate signal events that peak in M_{bc} from the ones that are not (which are then categorized as background events), the events reconstructed in the signal sample are fitted. with a sum of Crystal Ball function and Argus for each cut value on the corresponding variable to optimize (as in Fig. 2.3).

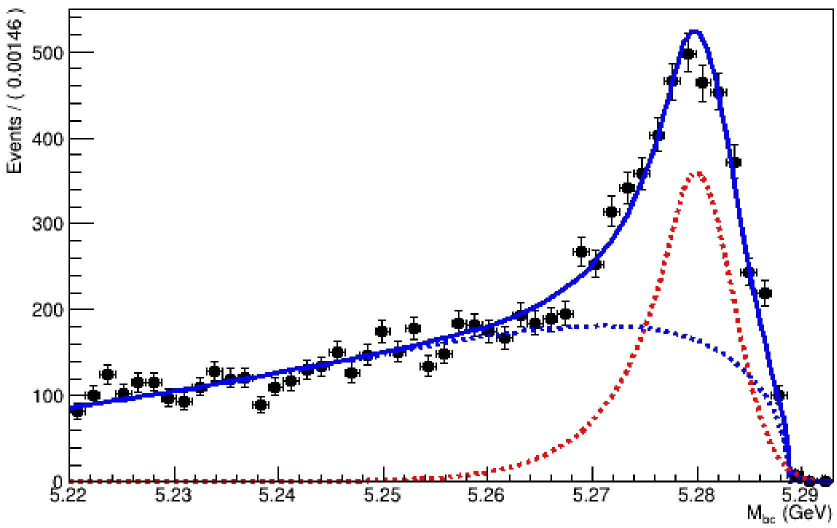


Figure (2.3) Example of a fit used to separate the correctly reconstructed B mesons (described by the red dotted Crystal Ball function) from the wrongly reconstructed ones (described by the blue dotted Argus function).

The next sections illustrate the procedure for each of the four decay channels.

2.4.1 $B^- \to \Lambda_c^+$ decays

Here below the procedure of optimized signal selection of charged correlated decays is presented.
First, in order to suppress the continuum background the cut on $foxWolframR2$ is optimized. Fig. 2.4 shows the $foxWolframR2$ distributions for signal and continuum events.

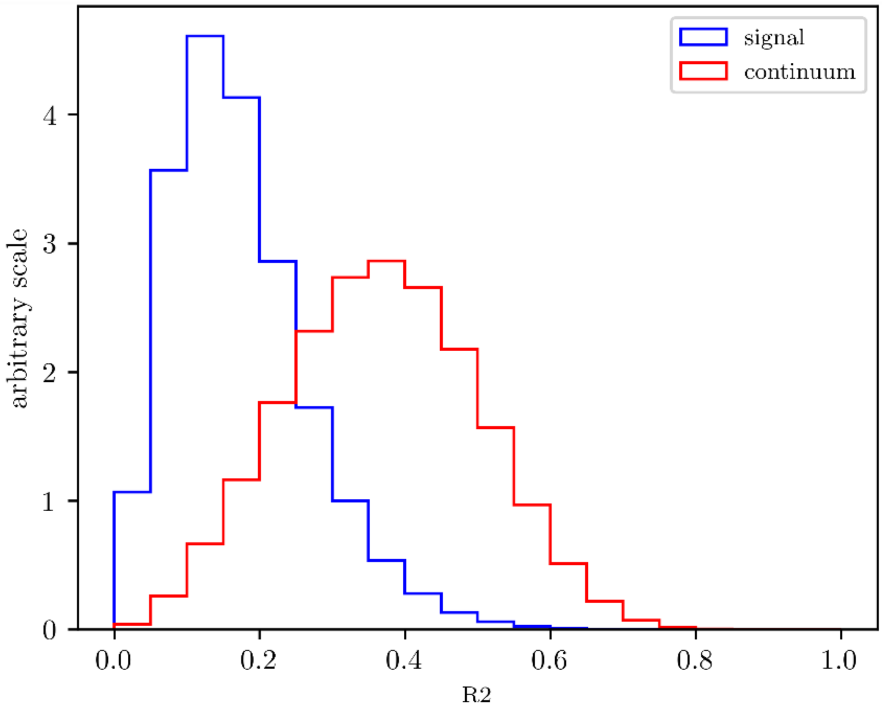


Figure (2.4) Distribution of the $foxWolframR2$ variable for signal and continuum background events.

With the optimized cut $foxWolframR2 < 0.27$ (corresponding to the maximum of the FOM curve shown in Fig. 2.5), the cut on SignalProbability is optimized in the same way (see Fig. 2.6).

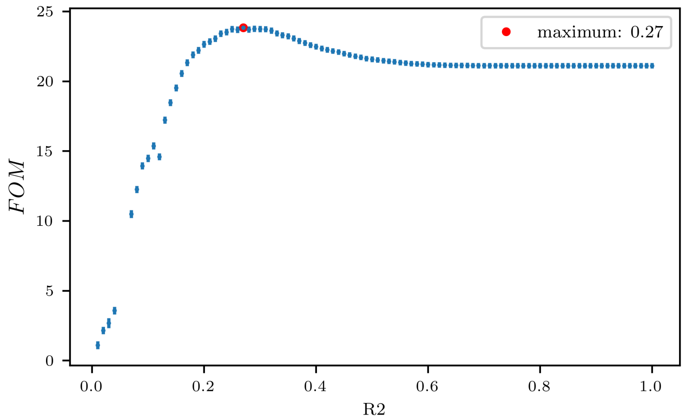


Figure (2.5) Figure of Merit values calculated at several cuts on the $foxWolframR2$ variable

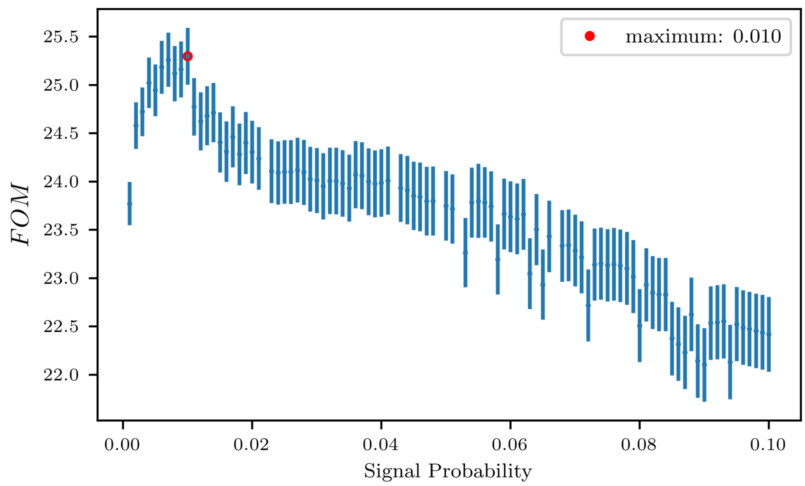


Figure (2.6) Figure of Merit values calculated at several cuts on the SignalProbability variable

With the optimized cut SignalProbability > 0.01, the cut on $foxWolframR2$ variable is rechecked (Fig. 2.7). Being the maximum values fluctuating around $foxWolframR2 < 0.3$, this cut is the one finally chosen for this variable.

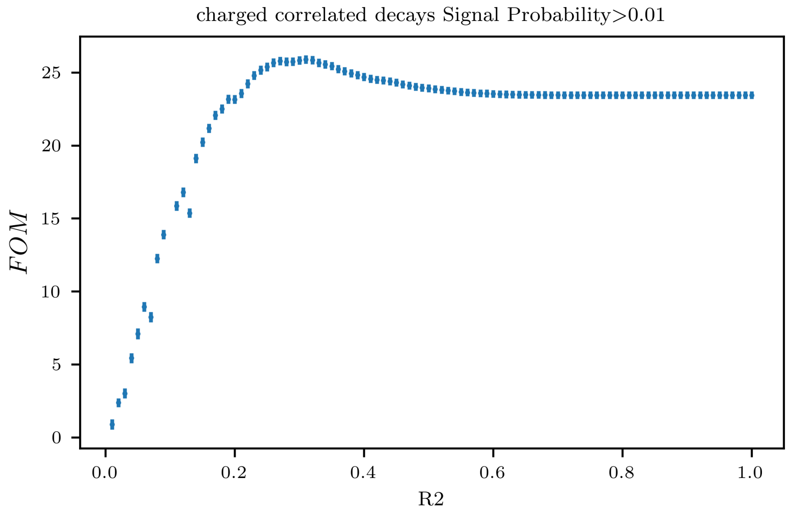


Figure (2.7) Figure of Merit values calculated at several cuts on the $foxWolframR2$ variable

With the optimized cuts on SignalProbability and $foxWolframR2$ variable, the cut on $p_{CMS}^{\Lambda_c}$ is optimized

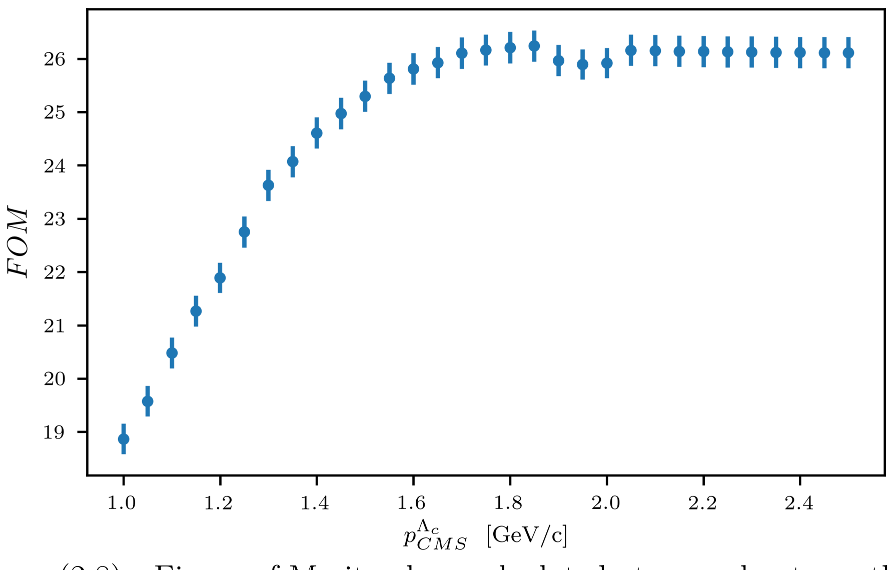


Figure (2.8) Figure of Merit values calculated at several cuts on the momentum of the Λ_c candidates in the center of mass system

From Fig. 2.8 one can see that with values of the cut above $p_{CMS}^{\Lambda_c} < 1.8$ GeV/c^2 a plateau of maximum FOM values is reached. But such a cut would still be useful to reject

some background events as one can see from Fig. 2.9.
Finally the optimized selection cuts are:

- $foxWolframR2 < 0.3$
- SignalProbability > 0.01
- $p_{CMS}^{\Lambda_c} < 1.8$ GeV/c

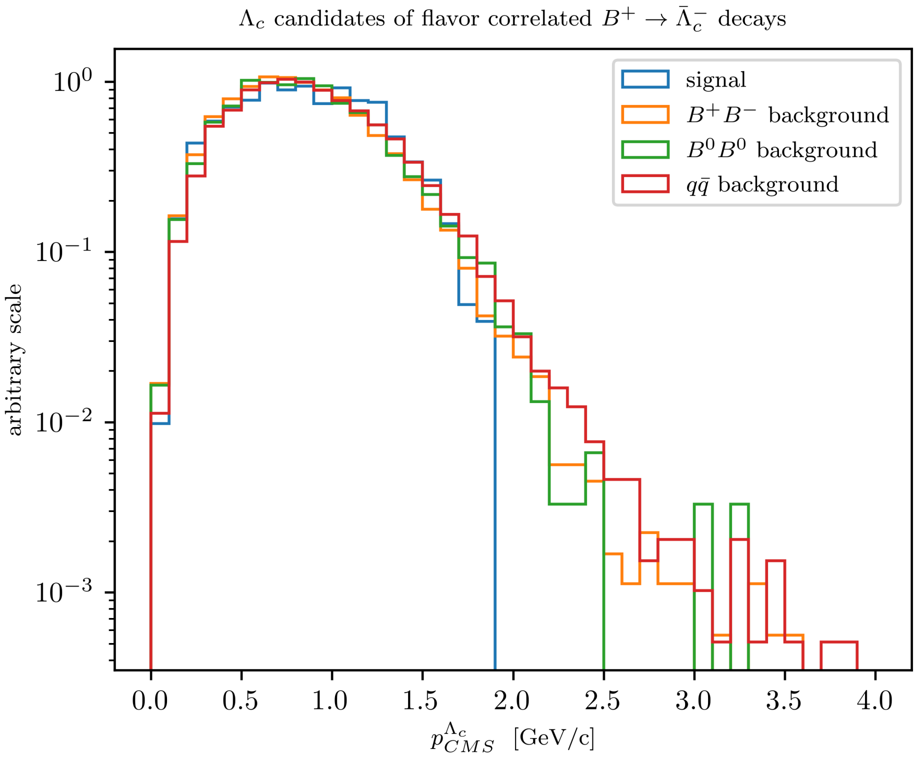


Figure (2.9) Distribution of Λ_c candidates momenta in the center of mass system

2.4.2 $B^+ \to \Lambda_c^+$ decays

For anticorrelated decays, the same $foxWolframR2$ cut is defined after the optimization procedure (see Fig. 2.13).

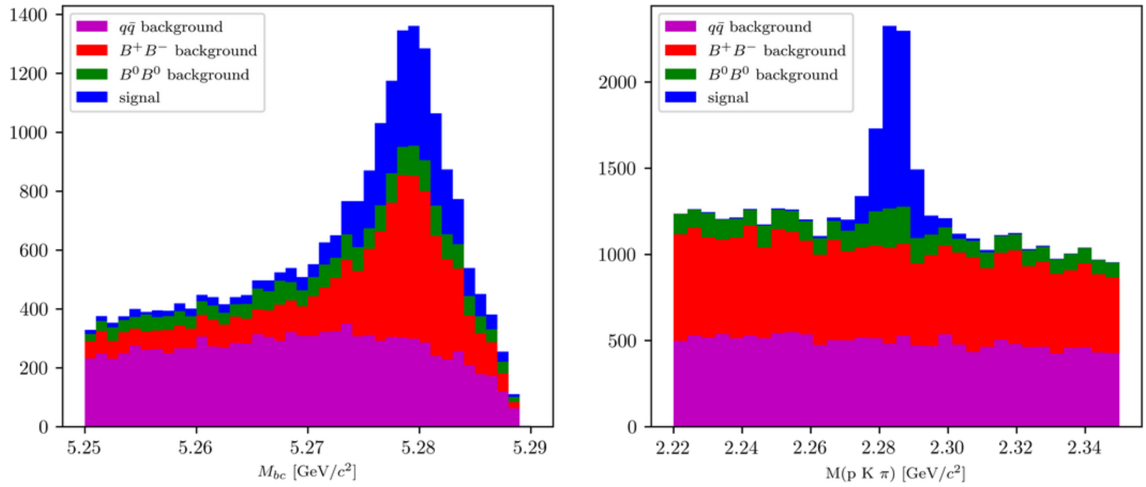


Figure (2.10) Distribution of M_{bc} (left) and invariant mass of charged correlated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

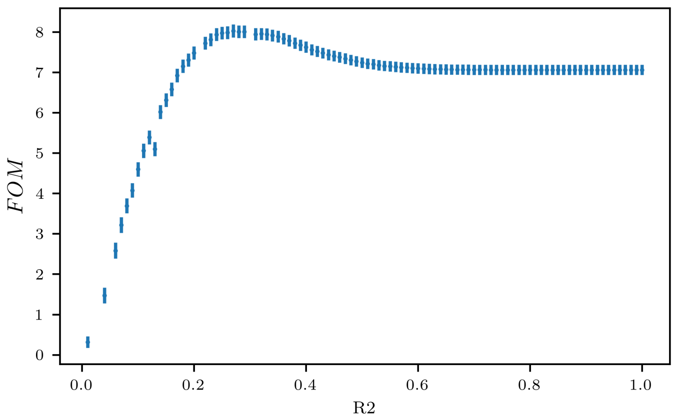


Figure (2.11) Figure of Merit values calculated at several cuts on the $foxWolframR2$ variable

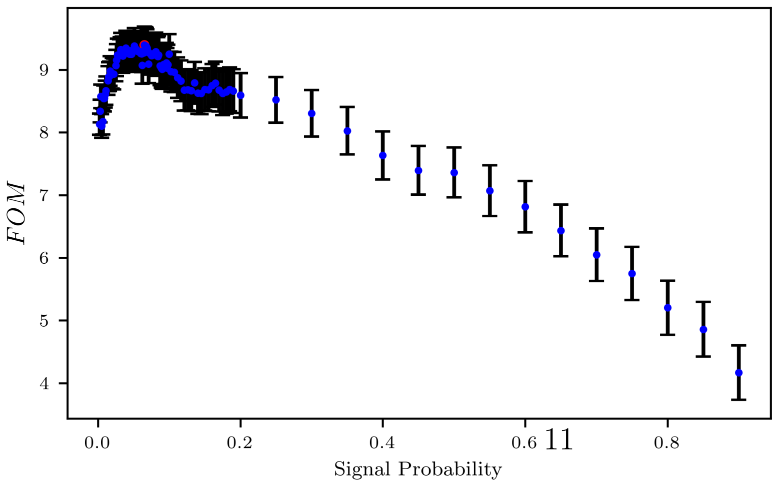


Figure (2.12) Figure of Merit values calculated at several cuts on the SignalProbability variable

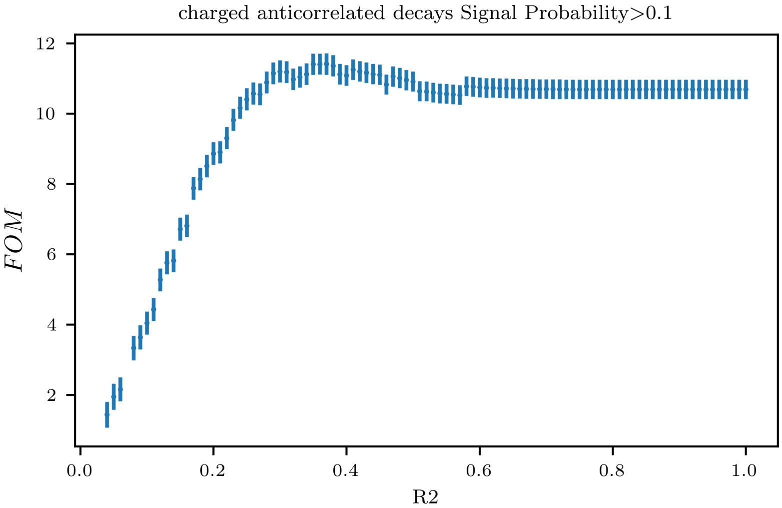


Figure (2.13) Figure of Merit values calculated at several cuts on the $foxWolframR2$ variable

With the optimized cuts on SignalProbability and $foxWolframR2$ variable, the cut on $p_{CMS}^{\Lambda_c}$ is selected.

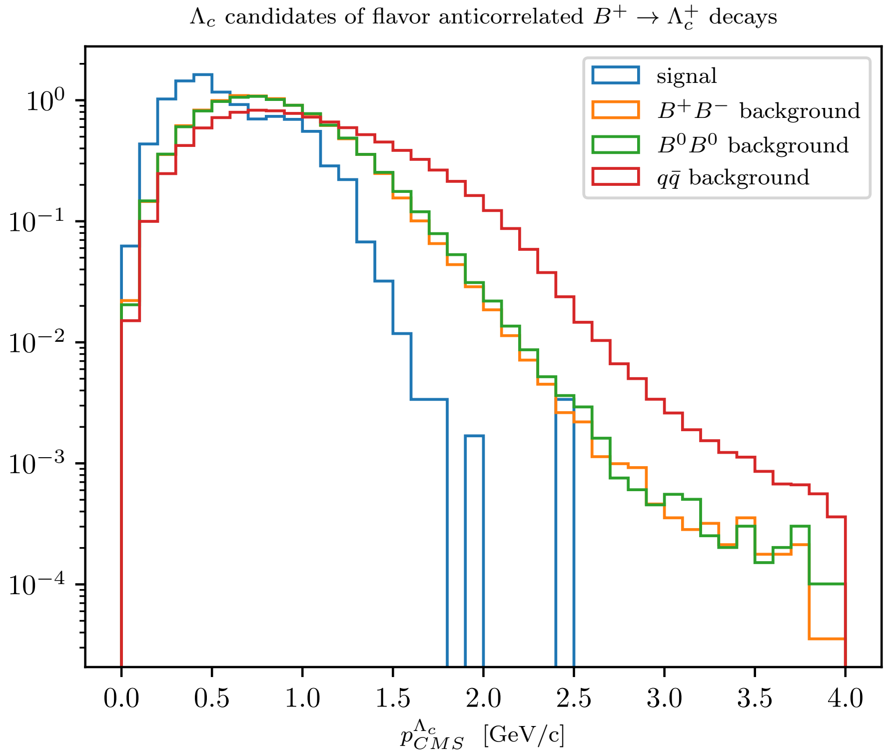


Figure (2.14) Distribution of Λ_c candidates momenta in the center of mass system

The final optimized selection cuts are:

- $foxWolframR2 < 0.3$
- SignalProbability > 0.1
- $p_{CMS}^{\Lambda_c} < 1.5$ GeV$/c$

Fig. 2.15 shows the projections of the M_{bc} and $M(pK\pi)$ distributions.

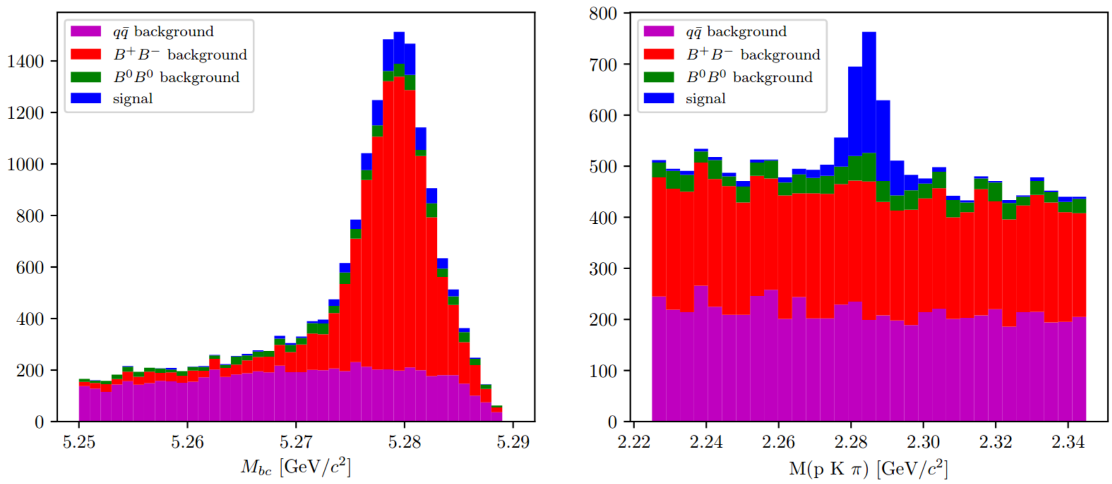


Figure (2.15) Distribution of M_{bc} (left) and invariant mass of charged correlated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

2.4.3 $\bar{B}^0 \to \Lambda_c^+$ decays

Also for neutral correlated decays, first the cut on $foxWolframR2$ is optimized.

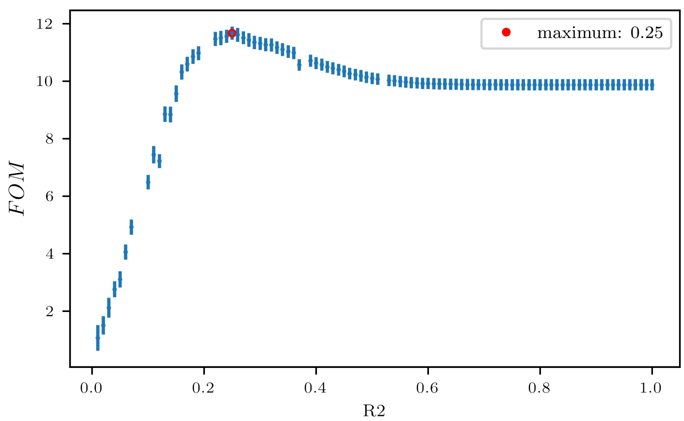


Figure (2.16) Figure of Merit values calculated at several cuts on the $foxWolframR2$ variable

The cut suggested by Fig. 2.16, $foxWolframR2 < 0.27$, is used to define the cut on the SignalProbability variable. From Fig. 2.17, it seems the optimal cut maximazing the FOM would be around 0.05. If one zooms like in Fig. 2.18 one can see that there is a sort of plateau starting around 0.03 and ending after values around 0.11, where the values fluctuate within statistical uncertainties around $FOM = 13$. In this case, a legitimate choice is to use the most stringent cut (SignalProbability > 0.11) being the FOM same but rejecting more background events.

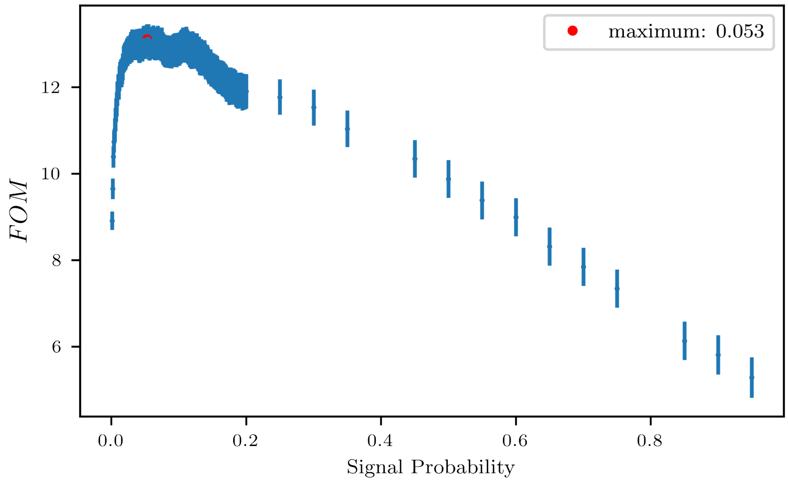


Figure (2.17) Figure of Merit values calculated at several cuts on the SignalProbability variable

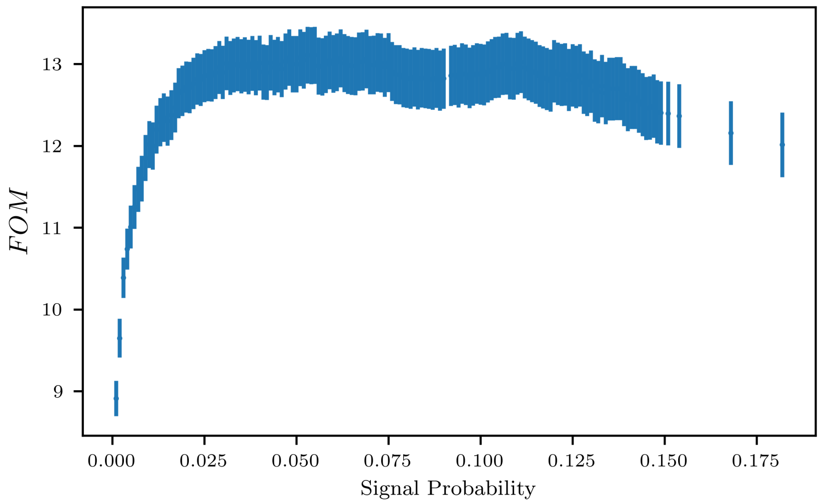


Figure (2.18) Figure of Merit values calculated at several cuts on the SignalProbability variable

The FOM curve for the $foxWolframR2$ variable is rechecked applying the chosen cut on SignalProbability (Fig. 2.19). As done in the other cases the final cut chosen is $foxWolframR2 < 0.3$

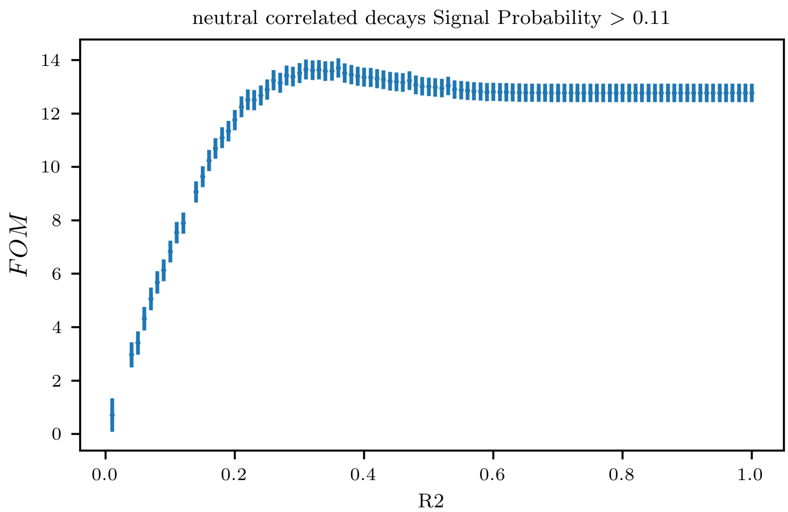


Figure (2.19) Figure of Merit values calculated at several cuts on the $foxWolframR2$ variable

Using now the two optimized cuts to check the FOM curve for the momenta a plateau appears for cuts starting from values around $p_{CMS}^{\Lambda_c} < 1.7$ GeV/c (see Fig. 2.20). In fact, around that value the level of background becomes signifcantly higher compared to the amount o signal events as one can see in Fig. 2.21

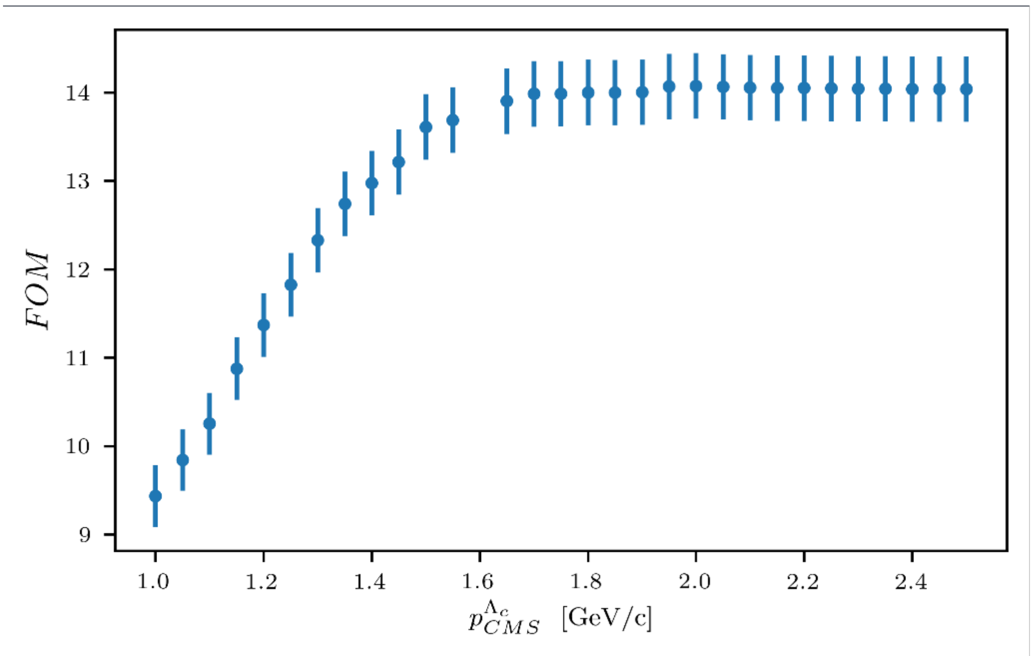


Figure (2.20) Figure of Merit values calculated at several cuts on the momentum of the Λ_c candidates in the center of mass system

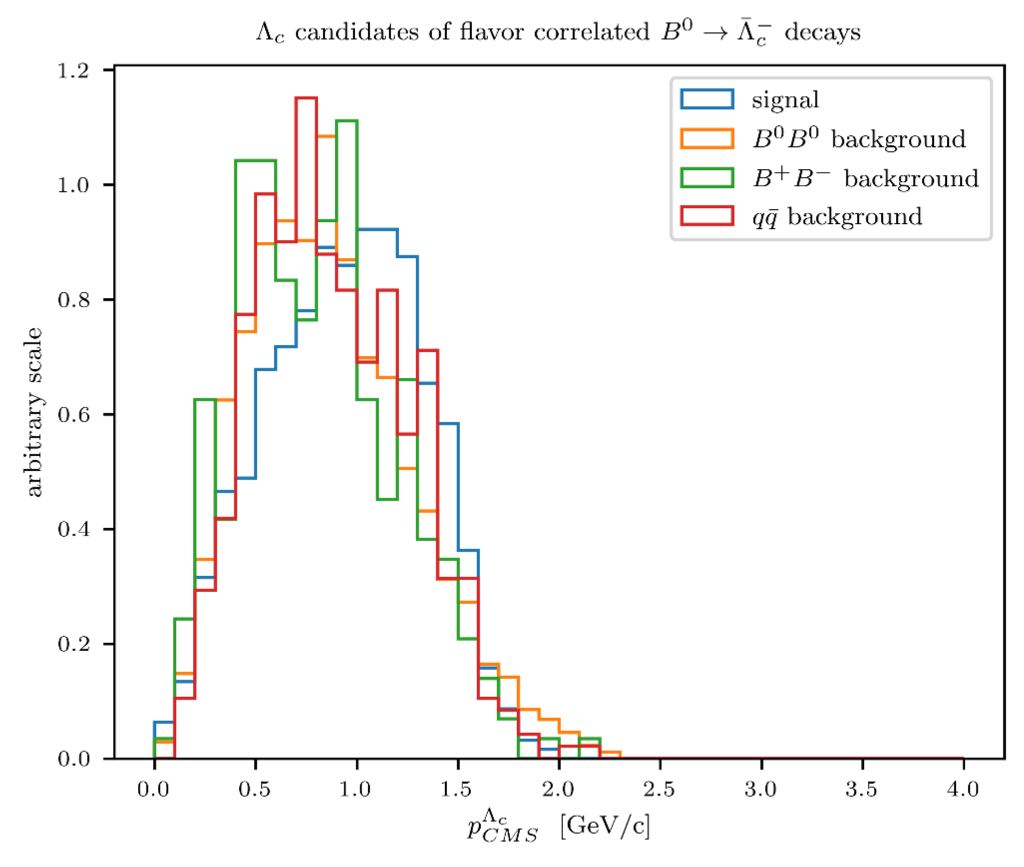


Figure (2.21) Distribution of Λ_c candidates momenta in the center of mass system

Fig. 2.22 shows the M_{bc} and $M(pK\pi)$ distributions after applying the following set of cuts:

- $foxWolframR2 > 0.3$
- SignalProbability > 0.11
- $p_{CMS}^{\Lambda_c} < 1.7$ GeV/c

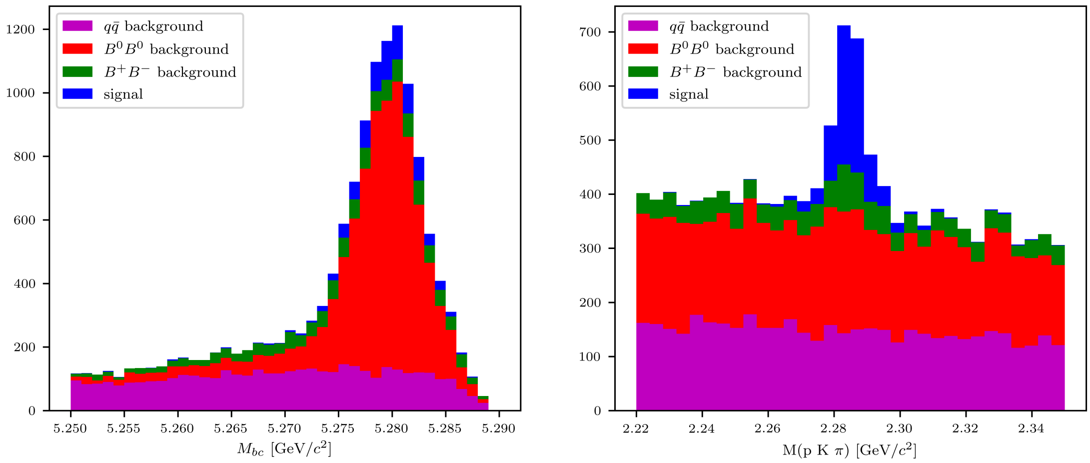


Figure (2.22) Distribution of M_{bc} (left) and invariant mass of neutral correlated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

2.4.4 $B^0 \to \Lambda_c^+$ decays

Finally same procedure is applied also to the neutral anticorrelated decays. The final selections for the variables of $foxWolframR2$, SignalProbability and the momentum of the Λ_c candidates in the center of mass system:

- $foxWolframR2 < 0.3$
- SignalProbability > 0.15
- $p_{CMS}^{\Lambda_c} < 1.4$ GeV/c

Fig. 2.26 shows the M_{bc} and $M(pK\pi)$ distributions after applying these cuts.

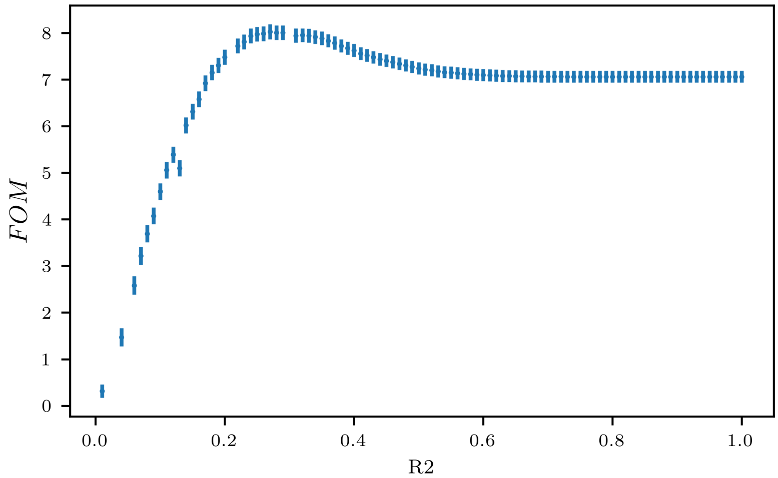


Figure (2.23) Figure of Merit values calculated at several cuts on the $foxWolframR2$ variable

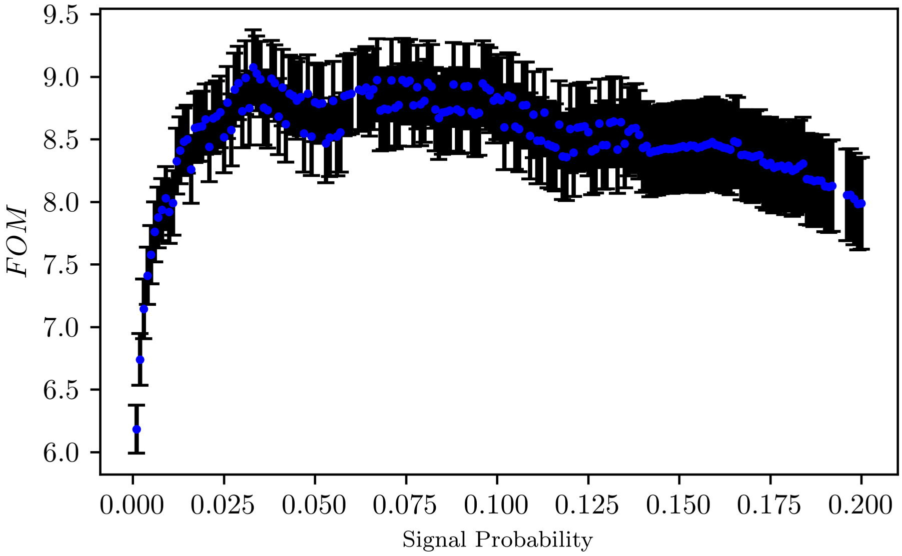


Figure (2.24) Figure of Merit values calculated at several cuts on the SignalProbability variable

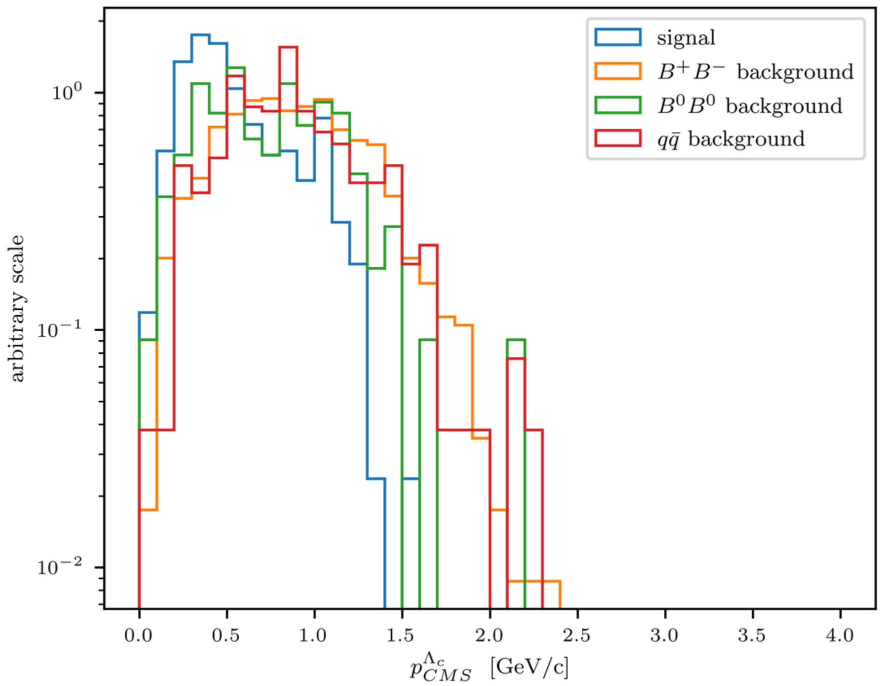


Figure (2.25) Distribution of Λ_c candidates momenta in the center of mass system

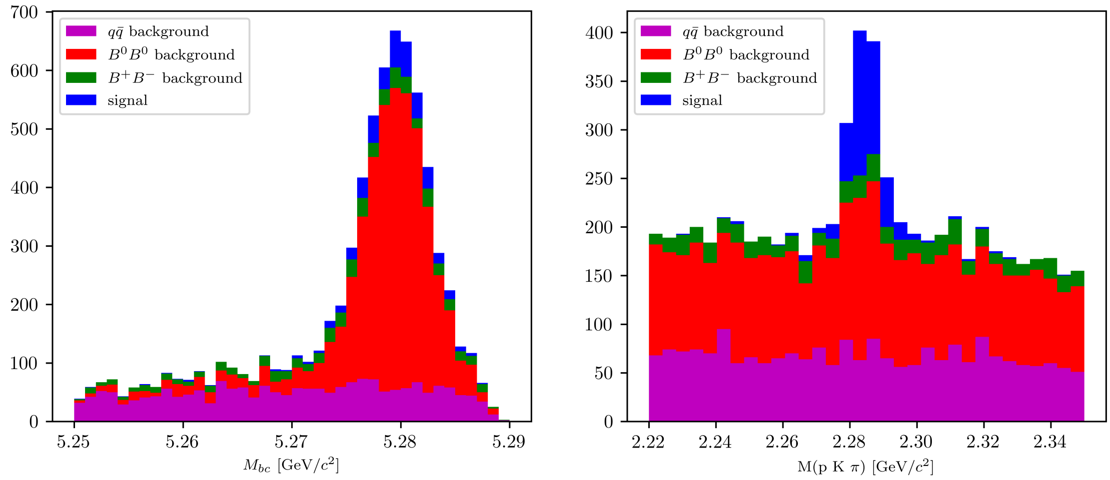


Figure (2.26) Distribution of M_{bc} (left) and invariant mass of neutral anticorrelated Λ_c candidates (right), in the signal region after the above mentioned selection cuts.

Chapter 3

2D simultaneous fit

3.1 Probability Density Functions (PDFs) for the two dimensional fit

The reconstructed events can be categorized as follows:

- peaking in both M_{bc} and $M(pK\pi\ )$
- peaking in M_{bc} but not in $M(pK\pi\ )$
- peaking in $M(pK\pi)$ but not in M_{bc}
- flat in both M_{bc} and $M(pK\pi\ )$

The first category is represented by the reconstructed signal: signal events which are correctly reconstructed. The signal events which are misreconstructed fall into the third category. The sum of the two is the so called "total signal". The PDFs used to describe the total signal distributions are discussed first.

3.1.1 Total Signal fits

For all the decays, the final sample of total signal events presents a peak around the expected B meson mass and a tail at low M_{bc} values.

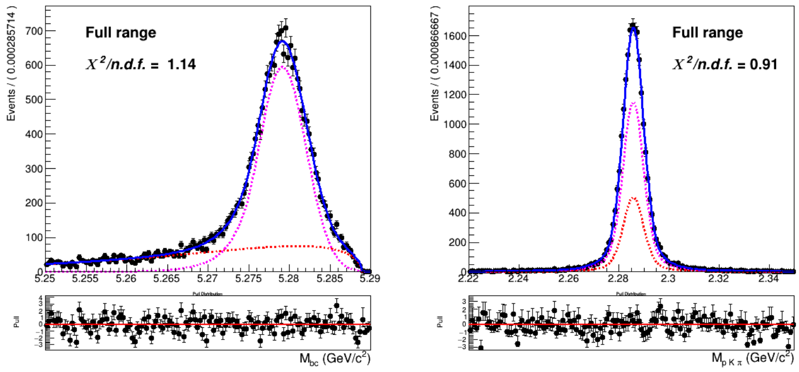


Figure (3.1) Two dimensional fit of charged correlated total signal events in M_{bc} and $M(pK\pi)$

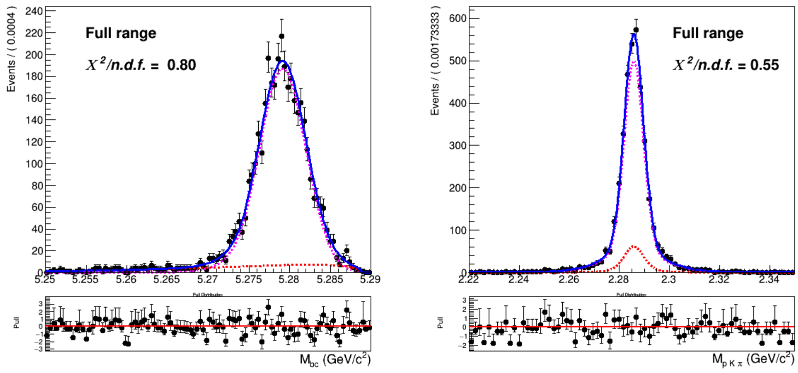


Figure (3.2) Two dimensional fit of charged anticorrelated total signal events in M_{bc} and $M(pK\pi)$

The 2D fits shown above are performed on five streams of signal MC with a sum of the following probability density functions:

$$P_{B,\Lambda_c}^{recSig}(M_{bc}, M(pK\pi)) = \Gamma_{CB}(M_{bc}) \times \rho_G(M(pK\pi)) \tag{3.1}$$

$$P_{B,\Lambda_c}^{misSig}(M_{bc}, M(pK\pi)) = \Gamma_{ARG}(M_{bc}) \times \rho_G(M(pK\pi)) \tag{3.2}$$

The first is used to fit the reconstructed signal and $\Gamma_{CB}(M_{bc})$ is a Crystal Ball function. The second is used to model the misreconstructed signal and $\Gamma_{ARG}(M_{bc})$ is an Argus function. In both cases a sum of three Gaussian functions $\rho_G(M(pK\pi))$ describes the mass of the Λ_c baryon.

As already said, only the events of reconstructed signal are considered as signal, while the misreconstructed signal is considered as background. [1]

[1] in the case of neutral decays, the fits shown are containing unmixed events only, the extracted parameters are then the same used to extract signal yields also when mixing is included.

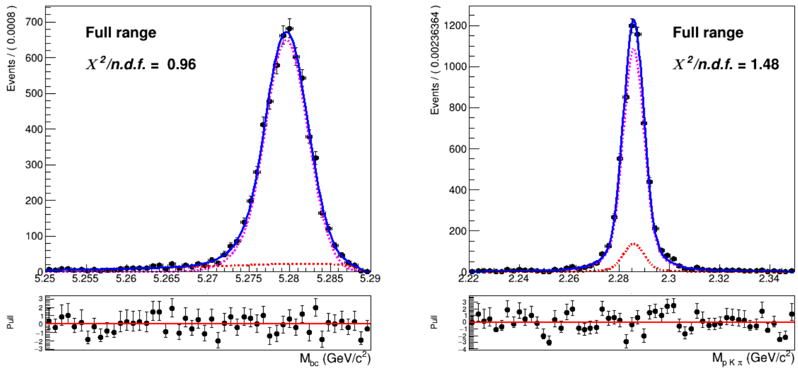


Figure (3.3) Two dimensional fit of neutral correlated total signal events in M_{bc} and $M(pK\pi)$

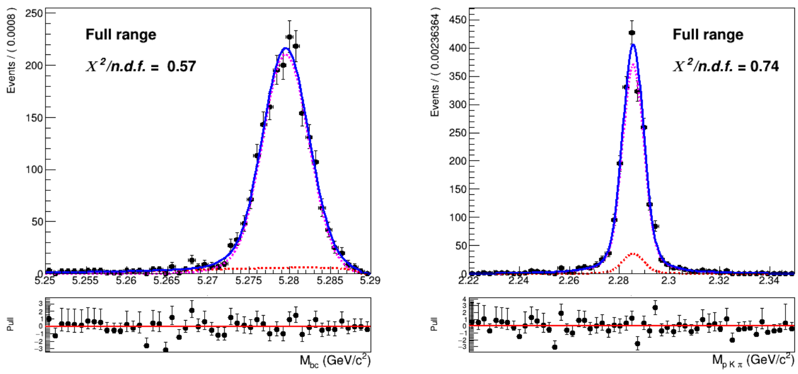


Figure (3.4) Two dimensional fit of neutral anticorrelated total signal events in M_{bc} and $M(pK\pi)$

3.1.2 Mbc peaking and flat background

The background composed of $B\bar{B}$ events where no Λ_c baryon is produced, is flat in its invariant mass, but can be distinguished in the following two categories:

- peaking in M_{bc} but not in $M(pK\pi)$
- flat in both M_{bc} and $M(pK\pi\ )$

as one can see in the following plots.

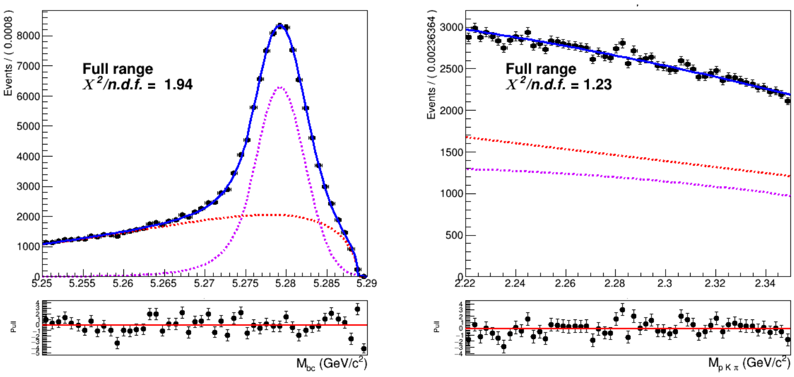


Figure (3.5) Two dimensional fit of charged correlated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

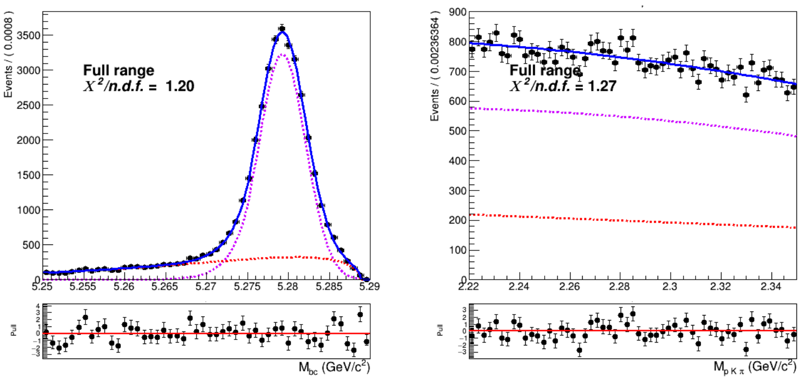


Figure (3.6) Two dimensional fit of charged anticorrelated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

This background presents a similar shape of the distribution in M_{bc}: the probability density functions used for it are again a Crystal Ball and an Argus.
The two types of background (peaking/flat in M_{bc}) are described by:

$$P^{peaking}_{B,\Lambda_c}(M_{bc}, M(pK\pi)) = \Gamma_{CB}(M_{bc}) \times \rho_{Cheb(a0,a1)}(M(pK\pi)) \tag{3.3}$$

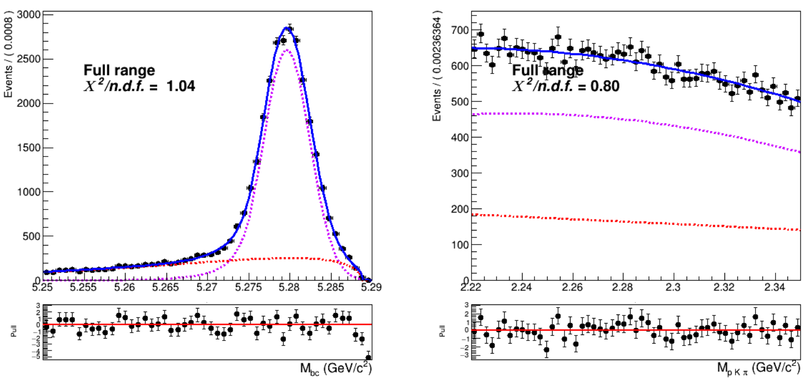


Figure (3.7) Two dimensional fit of neutral correlated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

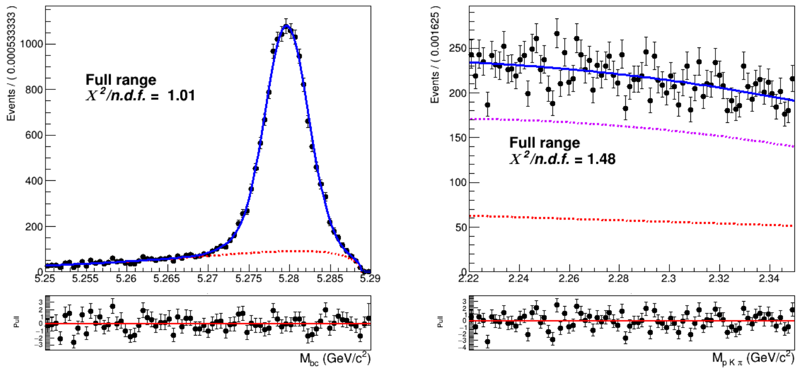


Figure (3.8) Two dimensional fit of neutral anticorrelated $B\bar{B}$ events in M_{bc} and $M(pK\pi)$

$$P^{flat}_{B,\Lambda_c}(M_{bc}, M(pK\pi)) = \Gamma_{ARG}(M_{bc}) \times \rho_{Cheb(b0)}(M(pK\pi)) \tag{3.4}$$

where $\rho_{Cheb(a0,a1)}(M(pK\pi))$ and $\rho_{Cheb(b0)}(M(pK\pi))$ represent a second order and first order Chebychev polynomial function respectively.

3.1.3 Crossfeed background

The contamination of misreconstructed $B^0 \to \Lambda_c$ events in the B^+ signal (and vice-versa) induces a background peaking in $M(pK\pi)$, but it also slightly peaks near the B meson mass, as one can see in Fig. 3.9

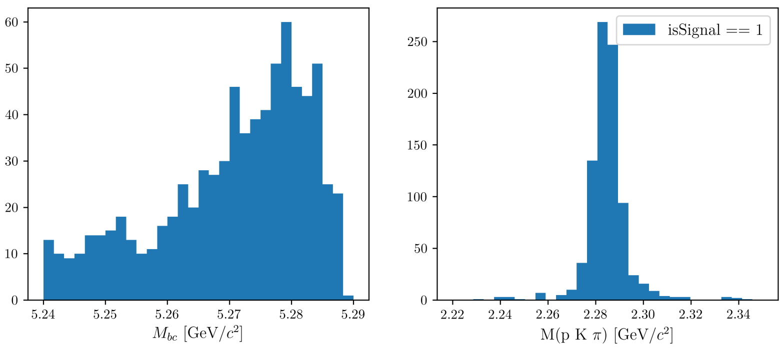


Figure (3.9) Crossfeed distribution in M_{bc} and $M(pK\pi)$

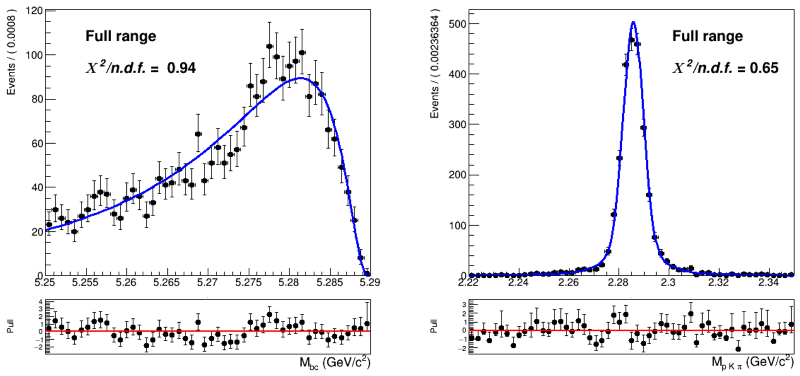


Figure (3.10) Two dimensional fit of crossfeed events in charged correlated channel in M_{bc} and $M(pK\pi)$

The 2D fit shown above are performed on five streams of signal MC with a sum of the

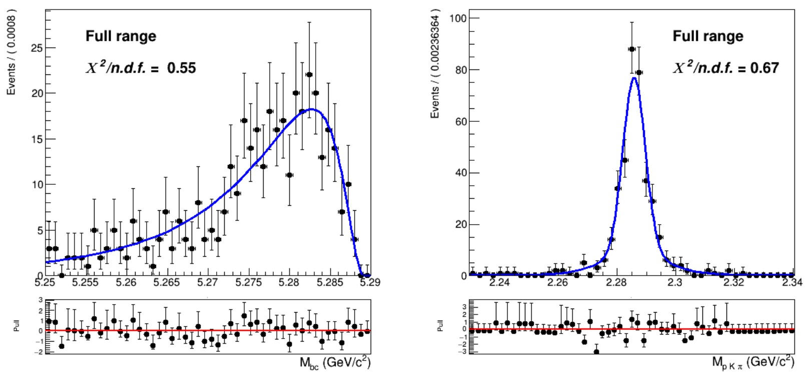


Figure (3.11) Two dimensional fit of crossfeed events in charged anticorrelated channel in M_{bc} and $M(pK\pi)$

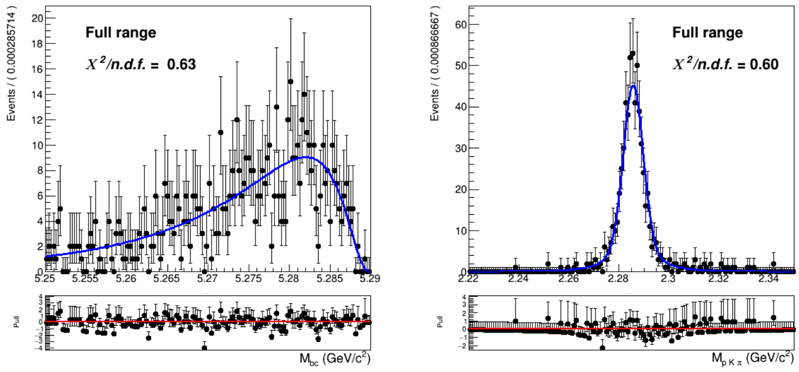


Figure (3.12) Two dimensional fit of crossfeed events in neutral correlated channel in M_{bc} and $M(pK\pi)$

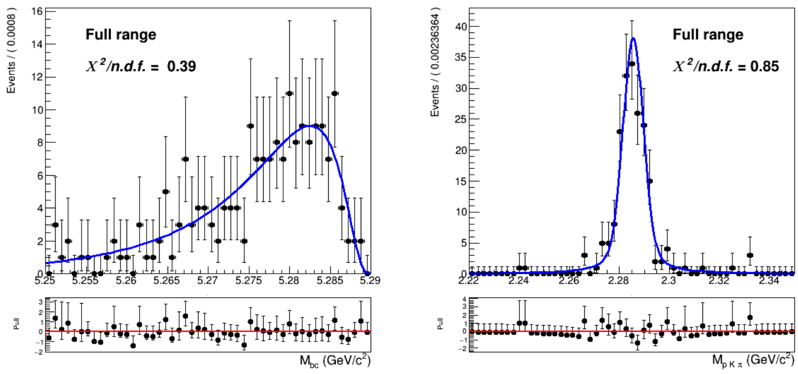


Figure (3.13) Two dimensional fit of crossfeed events in neutral anticorrelated channel in M_{bc} and $M(pK\pi)$

following probability density function:

$$P^{Crossfeed}_{B,\Lambda_c}(M_{bc}, M(pK\pi)) = \Gamma_{Novosibirsk}(M_{bc}) \times \rho_G(M(pK\pi)) \quad (3.5)$$

where $\Gamma_{Novosibirsk}(M_{bc})$ is a Novosibirsk function and the mass of the Λ_c baryon is described by the same sum of three Gaussian functions $\rho_G(M(pK\pi))$ as in Eq.3.1.

3.1.4 Continuum background

Besides the dataset recorded at the energy of the $\Upsilon(4S)$ resonance ($E^{on-res}_{CMS} = 10.58$ GeV), the *Belle* experiment recorded a sample of 89.4 fb^{-1} at an energy 60 MeV below the nominal $\Upsilon(4S)$ resonance ($E^{off-res}_{CMS} = 10.52$ GeV). The dataset allows to check for an appropriate modeling of the continuum MC simulation. Using the official tables (`https://belle.kek.jp/secured/nbb/nbb.html`) the off-resonance sample is scaled by

$$\frac{\mathcal{L}^{on-res}}{\mathcal{L}^{off-res}}\left(\frac{E_{CMS}^{off-res}}{E_{CMS}^{on-res}}\right)^2 \tag{3.6}$$

taking into account the difference in luminosity and in E_{CMS} (Energy in center of mass system).

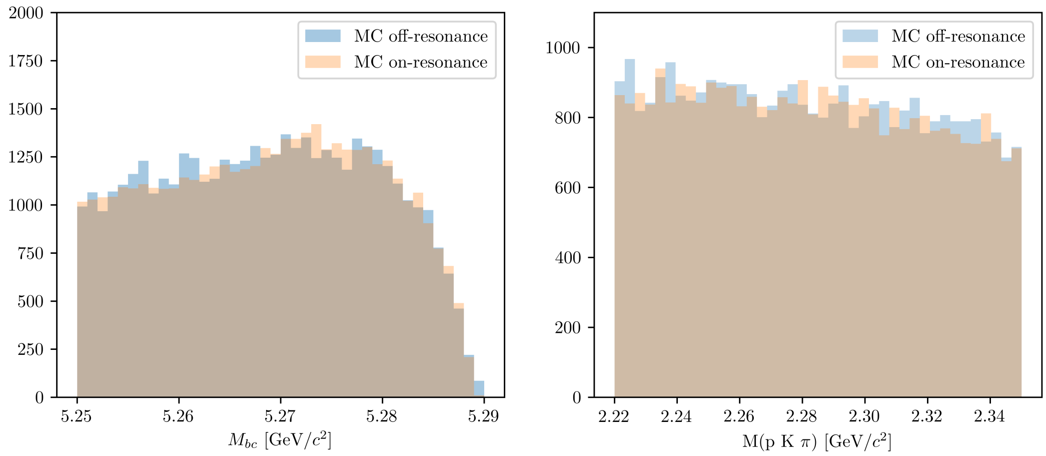


Figure (3.14) M_{bc} and $M(pK\pi)$ comparison between on-/off-resonance (scaled) Monte Carlo simulated continuum. The scaling is applied according to Eq. (3.6) and shifting the M_{bc} distribution by $E_{CMS}^{on-res} - E_{CMS}^{off-res}$.

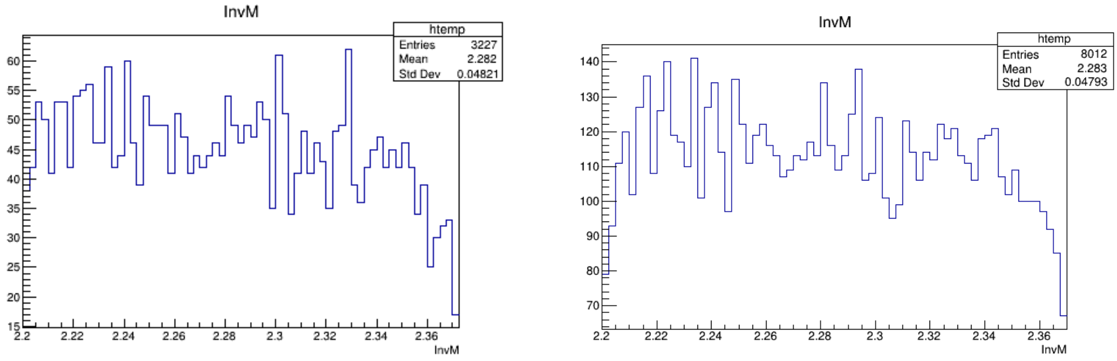
The plot in Fig.3.14 shows the M_{bc} and $M(pK\pi)$ distributions in the MC on-/off-resonance continuum after the scaling[2].

Ideally, provided that there's a good agreement between MC and data for the off-resonance sample and also between the MC on-/off-resonance continuum after the scaling, one could directly use the scaled off-resonance data to describe the continuum background in the fit on data. There are two reasons that prevent this very straightforward approach:

- First, since the off-resonance MC (and data) present very low statistics (Fig. 3.15a shows the Λ_c invariant mass in off-resonance data), scaling them with all the applied selection cuts would cause the PDF describing the continuum to be very much affected by statistical fluctuations.

- Secondly, the B meson candidates are reconstructed in both on-resonance and off-resonance events for values of $M_{bc} \geq 5.22$ GeV/c^2, but the E_{CMS} differs: there can be effects of correlations between the applied *SignalProbability* cut and the M_{bc} variable that one needs to take into account.

[2]it is obtained with the MC off-resonance sample being composed of 6 streams: the total amount is normalized



(a) (b)

Figure (3.15) On the left: Λ_c invariant mass in off-resonance data (all nominal cuts applied). On the right: Λ_c invariant mass in off-resonance data after the continuum suppression cut removal.

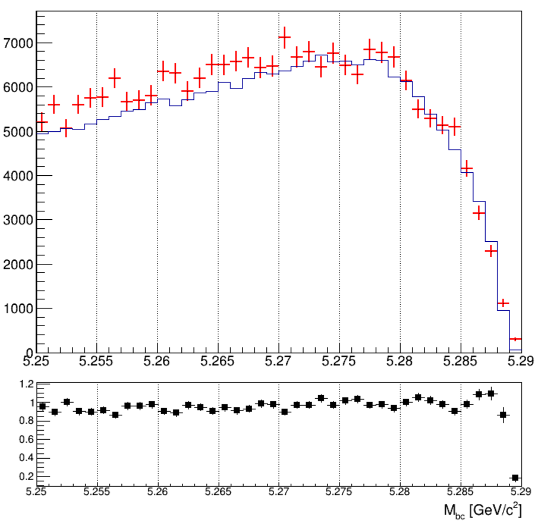
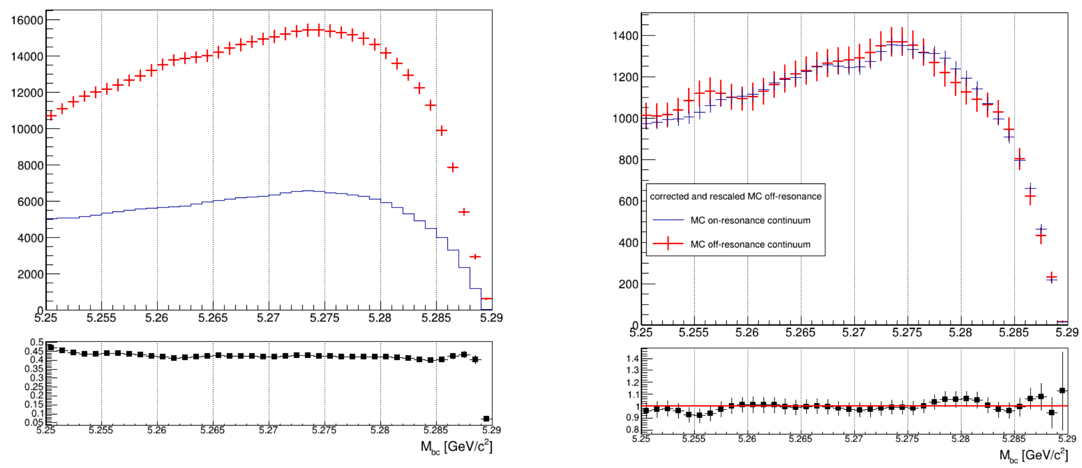


Figure (3.16) M_{bc} distributions of the MC (scaled) off-resonance sample (in red) and on-resonance (in blue) using 5 streams statistics and all nominal selection cuts applied.

In Fig. 3.16 one can notice some discrepancy in the shapes, apart from the not negligible statistical fluctuations in the (scaled) off-resonance distribution.



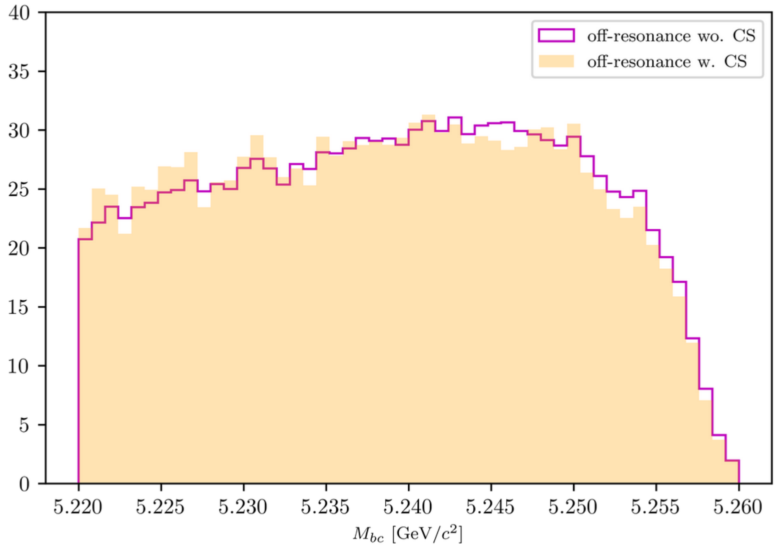
(a) (b)

Figure (3.17) On the left: M_{bc} distributions of the MC off-resonance sample without continuum suppression and the MC continuum sample with applied continuum suppression. On the right: M_{bc} distributions of the corrected scaled MC off-resonance and on-resonance MC continuum.

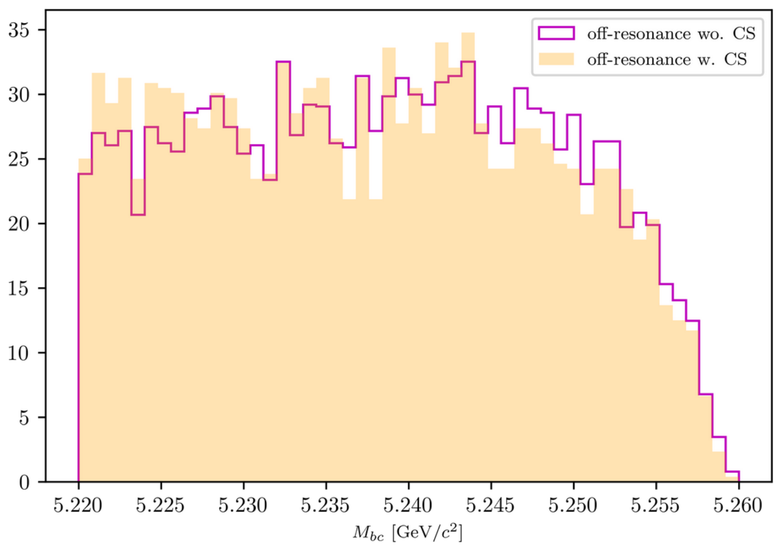
The procedure adopted to obtain the PDF describing the continuum background M_{bc} distribution is the following:

- 5 streams of off-resonance MC were scaled according to Eq. (3.6) without continuum suppression being applied and compared to the distribution of 5 streams of on-resonance continuum
- From a ratio plot, like the one in Fig. 3.17a, the bin-correction is obtained to correct the off-resonance data in the scaling procedure. To obtain the shape that can describe the continuum background M_{bc} distribution on data the continuum suppression is not applied on the off-resonance continuum sample, in order to acquire more statistics.

This procedure is first tested on an independent MC sample (see Fig. 3.17b) to check the result on simulated data before applying it on data.



(a)



(b)

Figure (3.18) Above: M_{bc} distributions of the MC off-resonance sample (5 streams) with and without continuum suppression. Below: M_{bc} distributions on data with and without continuum suppression.

The validity of the method relies on the fact that the difference between on-/off-resonance continuum events are well modeled in MC and that the shape of the M_{bc} distribution doesn't change significantly when removing the continuum suppression cut both on MC and data (as one can see from Figures 3.18a - 3.18b).
Additionally, the continuum suppression cut efficiency should be the same in data and MC in order to have the correct scaling on data with the above mentioned method. Fig.

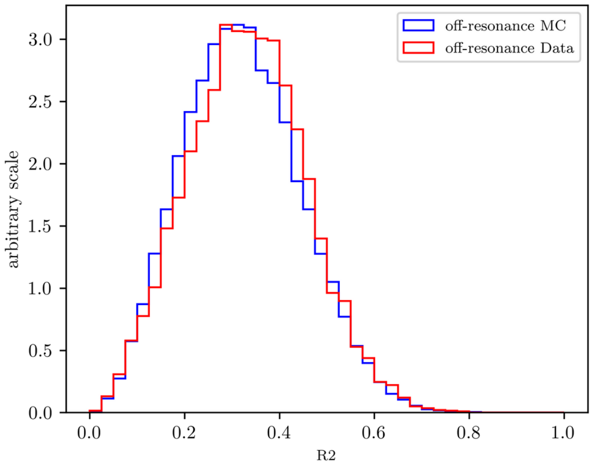


Figure (3.19) Distributions of variable $foxWolframR2$ in off-resonance MC and data.

3.19 shows the distribution of the $foxWolframR2$ variable in off-resonance MC and data. The slight shift visible in data can cause a different impact on data in terms of rejected continuum background when applying the $foxWolframR2 < 0.3$ cut. It is found to reject about 60% of the continuum background in data, whereas it rejects 55% of the continuum background in MC (56% in on-resonance MC). Therefore in data one can expect about 2.25% less continuum background events. This discrepancy is not statistically significant (the statistical uncertainty for the continuum background events is of the level of $\sim 1\%$), a simple correction to the number of events can be applied on data and its possible systematics can be then taken into account.
The obtained distribution can be then fitted with a Novosibirsk function (see Fig. 3.20). This is the procedure which can be then applied on the off-resonance data to obtain the M_{bc} shape describing the continuum background in data.

In the Λ_c invariant mass one doesn't expect correlation effects, but nevertheless there can be differences due to the limited statistics of the off-resonance sample. In fact, in the case of on-resonance MC for the charged correlated decays some events in which Λ_c candidates survive nominal selection cuts are visible and can be described with a small Gaussian on the top of the flat background (Fig.3.21a). Due to the low statistics one cannot see a similar peak in the off-resonance sample (the Fig.3.21b shows a 5 streams statistics).

The shape describing the Λ_c invariant mass in the case of charged correlated decays is obtained from the simulated on-resonance continuum, again using 5 streams statistics (see Fig. 3.21a). In the other cases no peak is visible and therefore also the Λ_c invariant mass shape is obtained from 5 streams of off-resonance events.

Finally, it is possible to examine the validity of the whole procedure on the independent stream. Fig. 3.22 - 3.25 show the M_{bc}, $M(pK\pi)$ projections of the overlayed two dimensional PDFs obtained with the above described procedure.
The 2D PDF used for Fig. 3.22 can be written as:

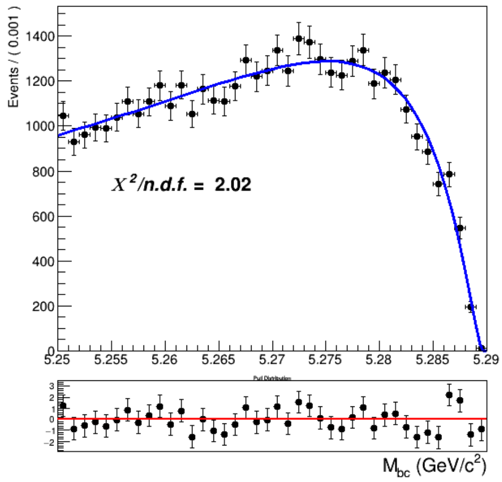
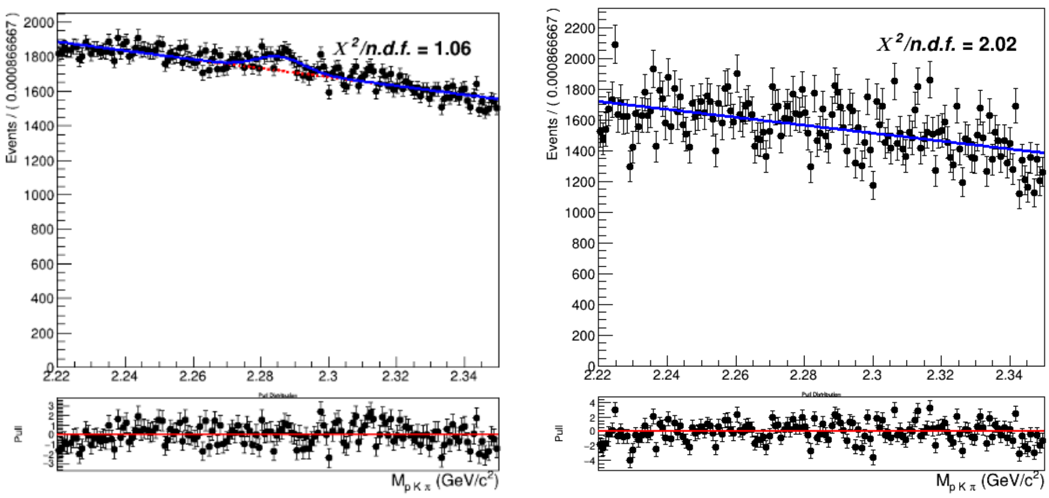


Figure (3.20) Fit of the M_{bc} distribution MC (scaled) off-resonance continuum (one stream).



(a) (b)

Figure (3.21) Comparison between 5 streams of MC on-resonance continuum 3.21a) and off-resonance (scaled) continuum in $M(pK\pi)$ (3.21b).

$$P_{B,\Lambda_c}^{Continuum}(M_{bc}, M(pK\pi)) = \Gamma_{Nov}(M_{bc}) \times [\rho_{Cheb1}(M(pK\pi) + \rho_G(M(pK\pi))]$$

where, as already anticipated, the invariant mass is described by a sum of a first order Chebychev polynomial and the peak by the same triple Gaussian PDF adopted for the signal. The 2D PDF used for Fig. 3.23 and Fig. 3.24 differ only in the invariant mass, not having the triple gaussian. Whereas the 2D PDF used for Fig. 3.25 can be written as:

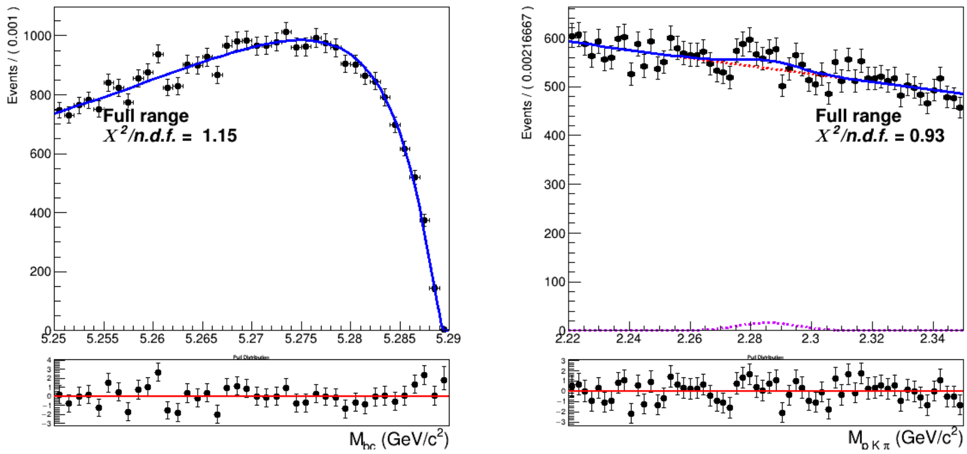


Figure (3.22) Two dimensional fit of charged correlated continuum events (one stream).

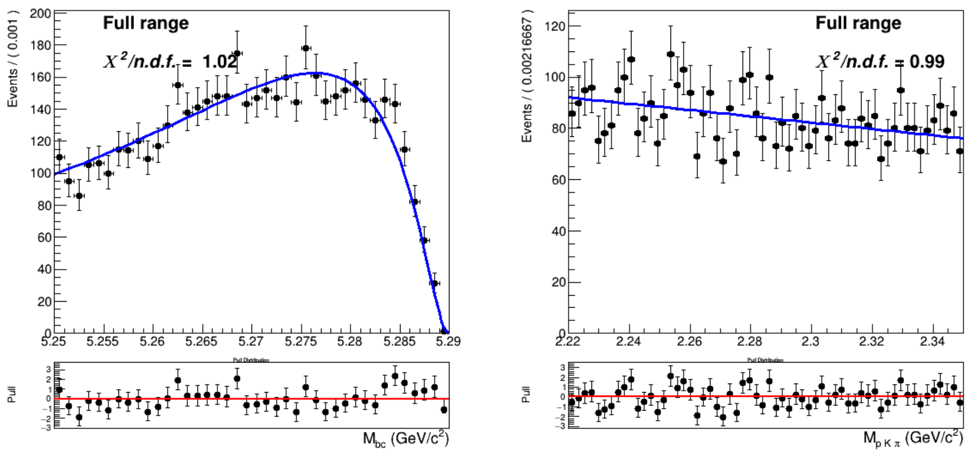


Figure (3.23) Two dimensional fit of charged anticorrelated continuum events (one stream).

$$P_{B,\Lambda_c}^{Continuum}(M_{bc}, M(pK\pi)) = \Gamma_{Argus}(M_{bc}) \times [\rho_{Cheb1}(M(pK\pi)]$$

since the distribution in M_{bc} can be better described by an Argus.

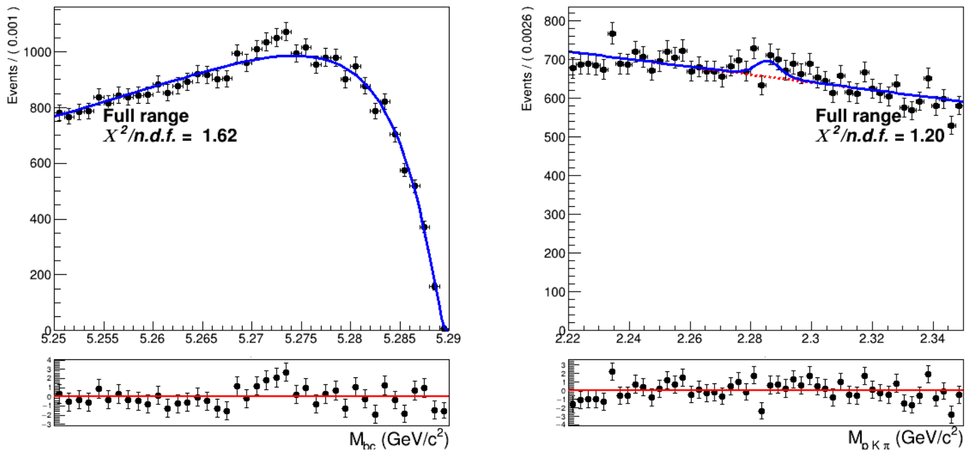


Figure (3.24) Two dimensional fit of continuum events (one stream).

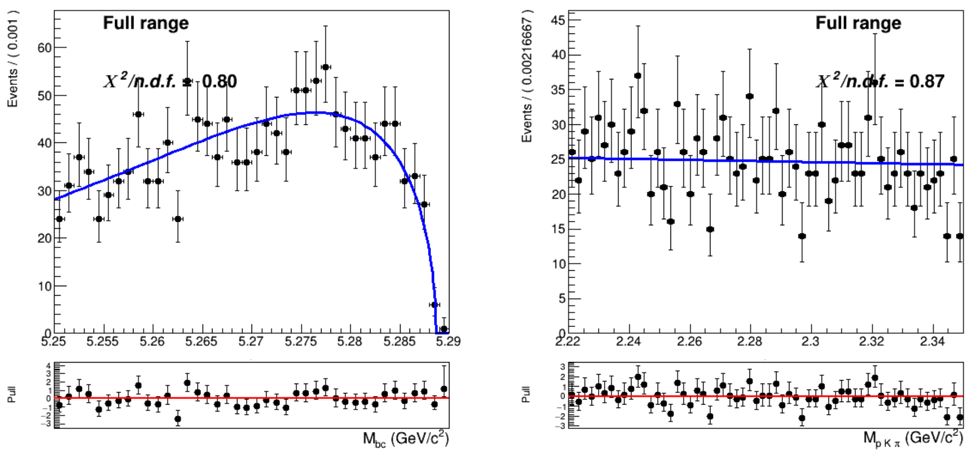


Figure (3.25) Two dimensional fit of continuum events (one stream).

3.2 Two dimensional fit on Monte Carlo

A total of six independent streams were used to construct/validate the fit model: each time five streams were used to construct the already discussed PDFs and the independent stream is used to test if the total PDF enables to extract the signal yield in an unbiased way (a total of six fits are performed on the six different streams of generic MC). To especially suppress the systematic uncertainties deriving from the amount of crossfeed the samples corresponding to the four different decay channels are fitted simultaneously. In all six fits all the shaping parameters are kept fixed, except:

- σ_{G1}: the width of the main of the three Gaussian functions in $\rho_G(M(pK\pi))$
- σ_{CB} parameter for the Crystal Ball describing the signal peak in M_{bc}

In the M_{bc} distribution the σ_{CB} width parameter for the Crystal Ball describing the M_{bc} peaking background is expressed as function of the signal σ_{CB} with a ratio fixed from MC. As for the normalizations, mis-/reconstructed signal events and M_{bc} peaking/flat background events are floated in the two dimensional unbinned maximum likelihood fits. The continuum background normalization is kept fixed to the value obtained by the off-resonance scaling procedure.

Instead, the normalization of crossfeed background events is determined by the fit in the following way:

$$N_{CrossBkg} = N^{corr}_{recSig} \cdot (\epsilon_{cross}/\epsilon_{recSig})^{corr} + N^{anticorr}_{recSig} \cdot (\epsilon_{cross}/\epsilon_{recSig})^{anticorr} \tag{3.7}$$

where N^{corr}_{recSig} ($N^{anticorr}_{recSig}$) are the fitted signal yields of the corresponding crossfeeding decay and $(\epsilon_{cross}/\epsilon_{recSig})$ are the ratios of misreconstruction efficiency and signal reconstruction efficiency as determined in the Monte Carlo. Since the crossfeed can likely occur both from correlated and anticorrelated channels, both are considered and summed together.

Exemplary, the distributions of stream 0 overlaid by the fitted PDF are depicted in Figs. 3.27 to 3.37. In Tables 3.1 to 3.4 the signal yields of the fits (**Reconstructed Signal**) to the two dimensional distributions for the six streams are listed and compared to the yields obtained from fits of signal distributions of each individual stream. The latter are the "expected" yields of reconstructed signal from a fit to the total signal events in the individual stream as the one plotted on Fig. 3.26 where all the parameters of the PDFs described in Eq. (3.1) are kept fixed and the corresponding yields are extracted from the fit. Note that in the case of Tables Table 3.3 - 3.4 the reconstructed signal yields are contaminated by events from the other neutral B^0 decay channel that experience mixing. Since those events cannot be analytically distinguished from the true signal events, the branching fraction calculation for neutral decays needs to take them into account and correct for their contribution (as will be shown in the next Chapter).

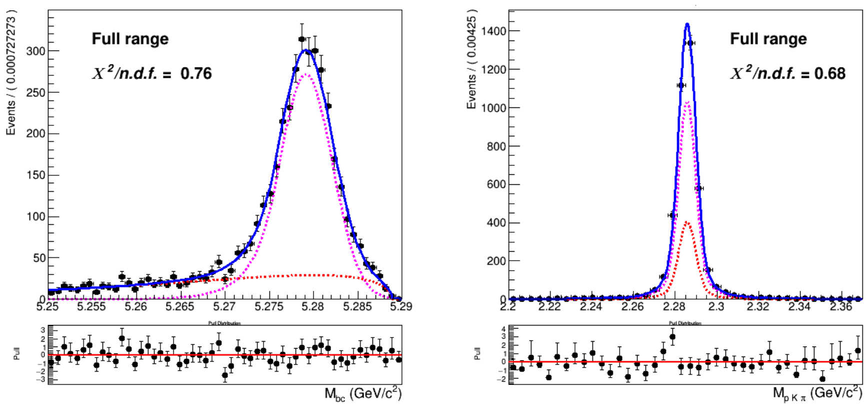


Figure (3.26) Two dimensional fit of Total Signal of stream 0 used to extract the expected reconstructed (corresponding to the PDF colored in magenta) and expected misreconstructed yields (corresponding to the PDF colored in red).

3.2.1 Fit results for $B^- \to \Lambda_c^+$ decays

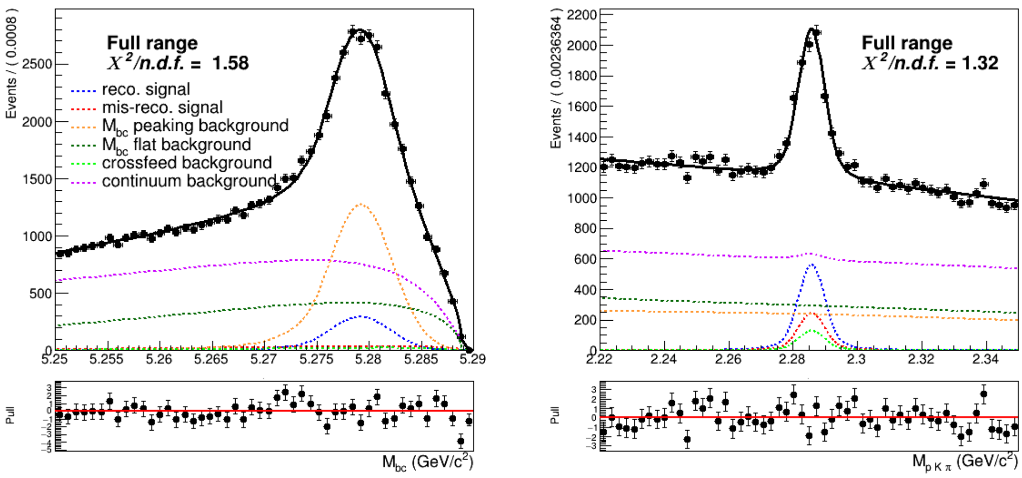


Figure (3.27) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged correlated decays.

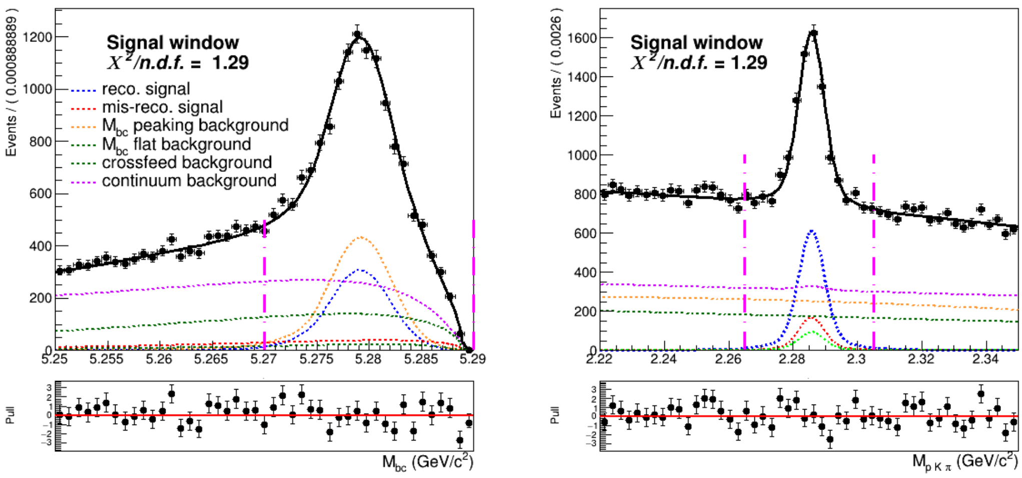


Figure (3.28) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged correlated decays(in the signal window)

	Reconstructed Signal		Total Signal			
	fit	expected	fit	MC truth	fit - MC truth	
stream 0	2829 ± 130	2928 ± 66	4063 ± 157	4061	-2	-0.05%
stream 1	2811 ± 137	2956 ± 65	4095 ± 161	4084	11	0.3%
stream 2	2952 ± 141	2940 ± 65	4345 ± 165	4138	207	5.0%
stream 3	2747 ± 134	2867 ± 66	4267 ± 160	4105	162	3.9%
stream 4	3043 ± 138	3017 ± 67	4148 ± 157	4176	-28	-0.7%
stream 5	2818 ± 136	2816 ± 65	3999 ± 162	4001	-2	-0.05%
sum	17200	17524	24917	24565	348	1.4%

Table (3.1) Charged correlated decays: comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

The yields obtained by the fit show a slight tendency of underestimation, but when comparing the sums of them with the sum of expected values one can see the difference is within statistical fluctuations (within 2%).
The tendency of underestimation is visible in , but it is just slightly larger than σ

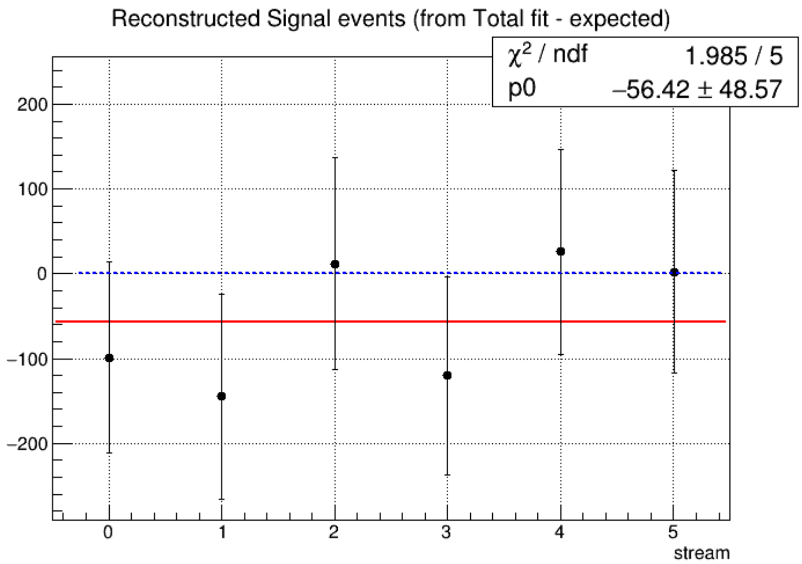


Figure (3.29) Residuals of signal yields (alias reconstructed signal, values reported in the first two columns of the above displayed table).

3.2.2 Fit results for $B^- \to \bar{\Lambda}_c^-$ decays

For the anticorrelated decays from charged B mesons the slight tendency is in the opposite direction. But the distribution of residuals (see Fig. 3.30) shows this slight overestimation is well within the statistical uncertainty.

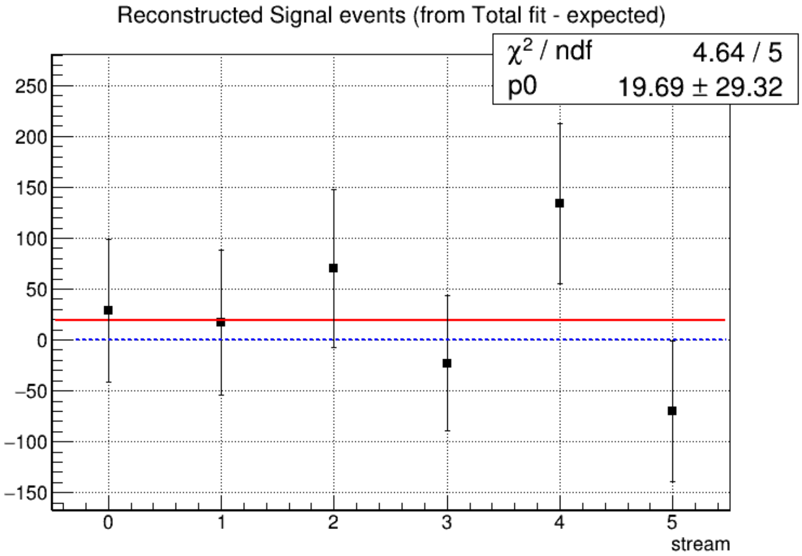


Figure (3.30) Residuals of signal yields (alias reconstructed signal, values reported in the first two columns of the above displayed table).

	Reconstructed Signal		Total Signal			
	fit	expected	fit	MC truth	fit - MC truth	
stream 0	724 ± 64	695 ± 28	813 ± 73	765	48	6.3%
stream 1	726 ± 65	709 ± 29	802 ± 67	785	17	2.2%
stream 2	788 ± 72	718 ± 29	911 ± 72	797	114	14.3%
stream 3	679 ± 60	702 ± 29	768 ± 66	802	-34	4.2%
stream 4	844 ± 73	710 ± 29	982 ± 73	785	197	25.1%
stream 5	605 ± 63	675 ± 29	722 ± 63	760	-38	-5.0%
sum	4366	4209	4998	4694	304	6.5%

Table (3.2) Charged anticorrelated decays: comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

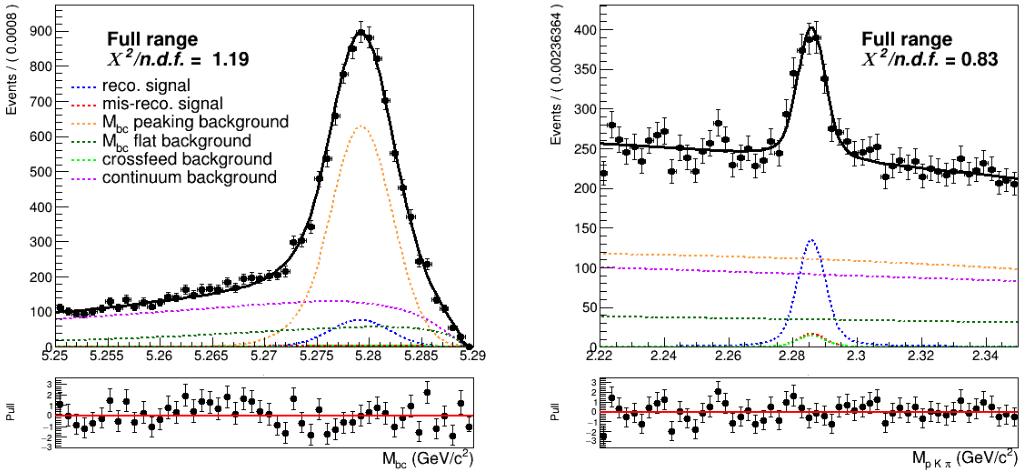


Figure (3.31) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged anticorrelated decays.

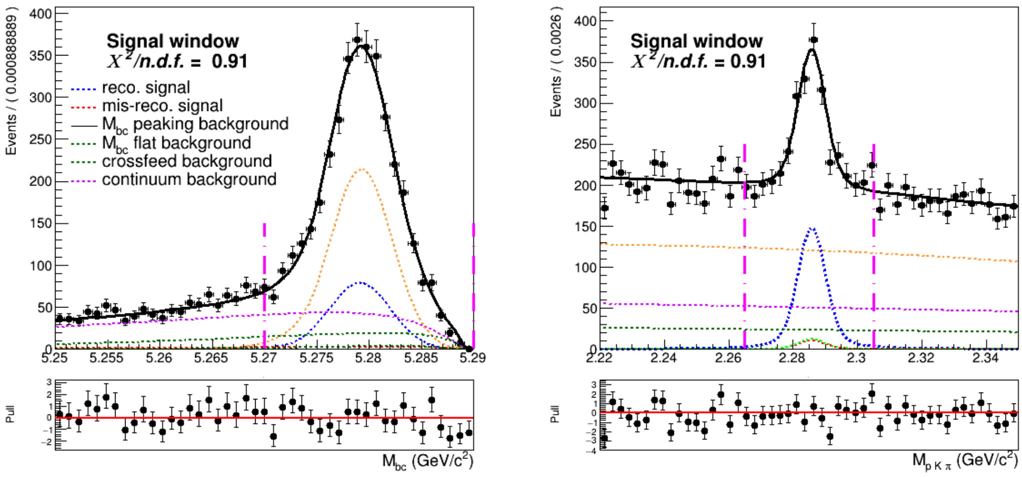


Figure (3.32) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for charged anticorrelated decays (in the signal window).

3.2.3 Fit results for $\bar{B}^0 \to \Lambda_c^+$ decays

Also in the case of the neutral correlated one can observe a slight overestimation in the reconstructed signal yields. The residuals (see Fig. 3.33) show that this is not so negligible: the significance is about 1.5σ.

	Reconstructed Signal		Total Signal			
	fit	expected	fit	MC truth	fit - MC truth	
stream 0	1329 ± 69	1240 ± 38	1390 ± 70	1379	11	0.8%
stream 1	1314 ± 68	1327 ± 38	1319 ± 66	1287	14	2.5%
stream 2	1266 ± 68	1254 ± 38	1273 ± 65	1251	22	1.8%
stream 3	1320 ± 67	1248 ± 38	1316 ± 66	1255	35	4.9%
stream 4	1227 ± 67	1214 ± 38	1300 ± 66	1223	77	6.3%
stream 5	1282 ± 70	1233 ± 38	1251 ± 66	1246	5	0.4%
sum	7738	7516	7709	7525	140	1.86%

Table (3.3) Comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

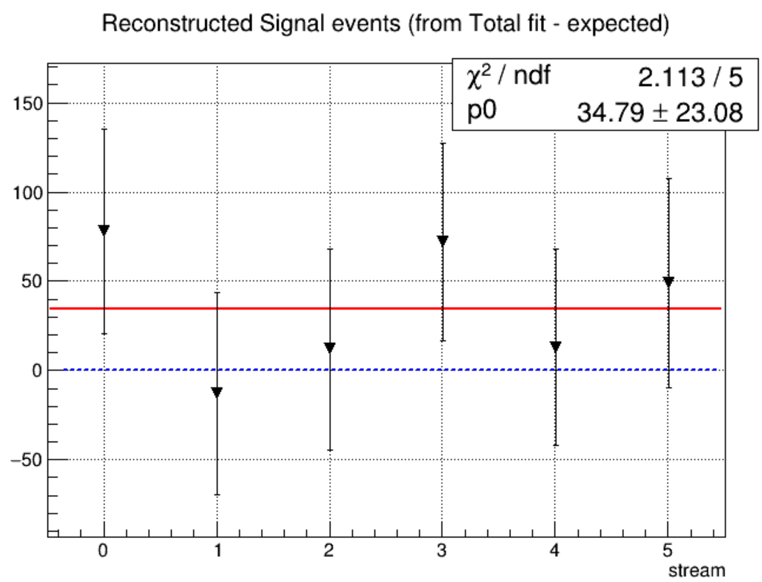


Figure (3.33) Residuals of signal yields for neutral correlated decays.

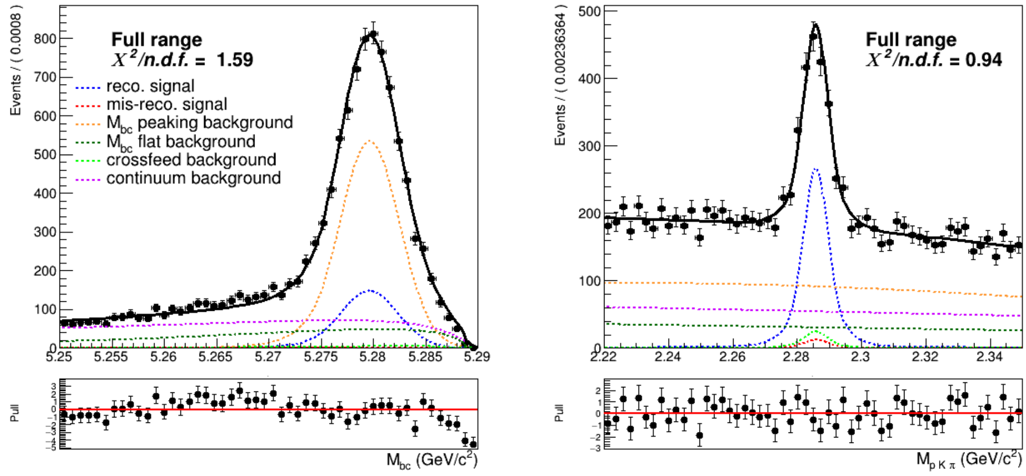


Figure (3.34) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral correlated decays.

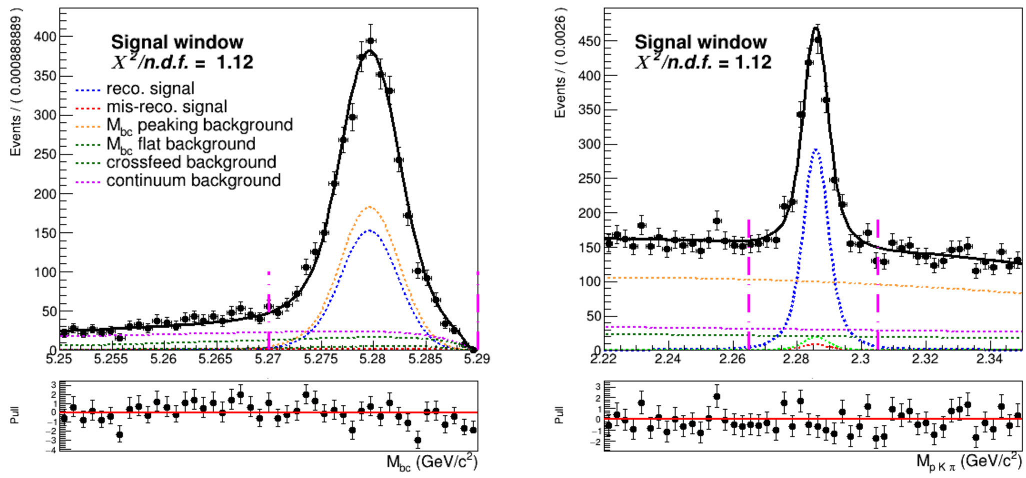


Figure (3.35) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral correlated decays.

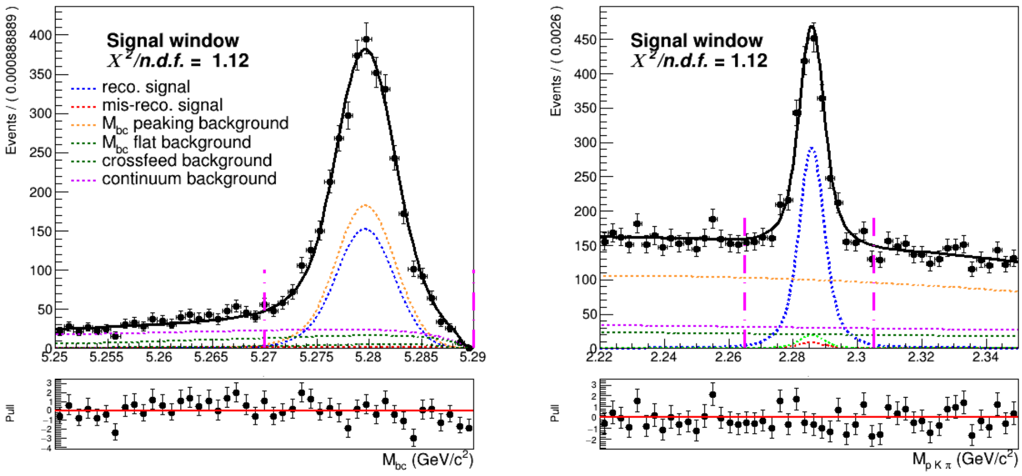


Figure (3.36) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral correlated decays (in the signal window).

3.2.4 Fit results for $\bar{B}^0 \to \bar{\Lambda}_c^-$ decays

	Reconstructed Signal		Total Signal			
	fit	expected	fit	MC truth	fit - MC truth	
stream 0	567 ± 45	575 ± 29	657 ± 45	668	-11	-1.6%
stream 1	565 ± 46	607 ± 21	660 ± 47	687	-27	-3.9%
stream 2	574 ± 46	593 ± 20	659 ± 45	639	20	3.1%
stream 3	714 ± 49	596 ± 21	792 ± 52	680	112	16.4%
stream 4	603 ± 50	594 ± 21	693 ± 50	638	55	8.6%
stream 5	589 ± 63	595 ± 21	603 ± 44	624	-21	3.4%
sum	3612	3560	4064	3936	170	4.3%

Table (3.4) Neutral anticorrelated decays: comparison of fitted and expected signal yields, fitted and truth-matched total signal for six streams of Belle generic MC when fitting the two dimensional distributions of M_{bc} and $M(pK\pi)$.

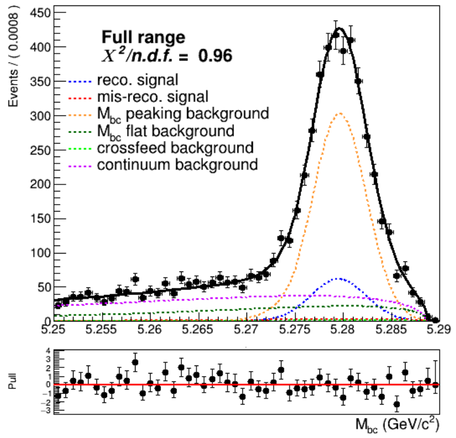


Figure (3.37) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral anticorrelated decays.

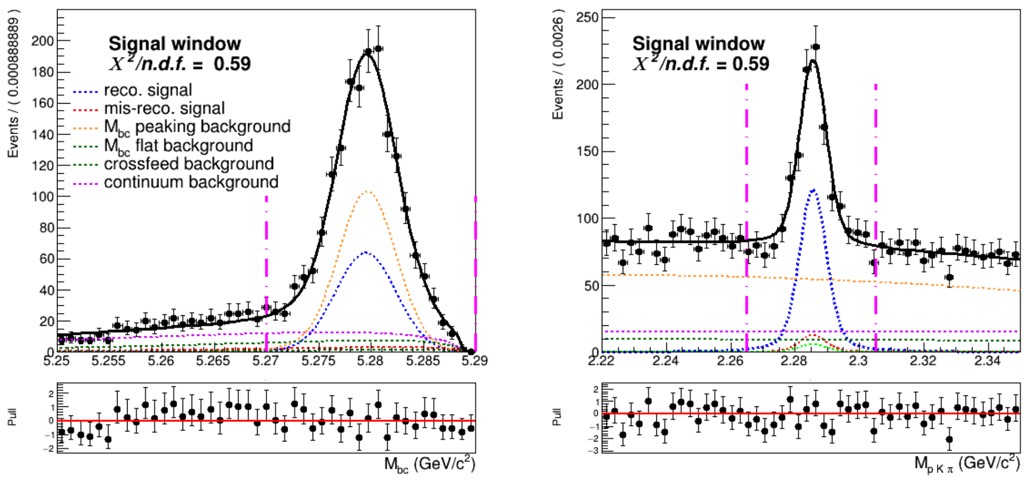


Figure (3.38) Projections of the two dimensional simultaneous fit on stream 0 Monte Carlo simulated data for neutral anticorrelated decays (in the signal window).

3.2.5 Fit residuals

Fig. 3.39 shows the residuals in each fit (stream by stream), calculated as the difference of reconstructed signal yield in the two dimensional fit from the expected value (in the fit of the total signal events). Since the signal events are the same in the two fits, the resulting yields are correlated. Therefore the uncertainties on the residuals are calculated (on a first approximation) as:

$$\sigma_{res} = \sqrt{(\sigma_{tot}^2 - \sigma_{sig}^2)} \tag{3.8}$$

Some of the results may seem not well distributed around 0.

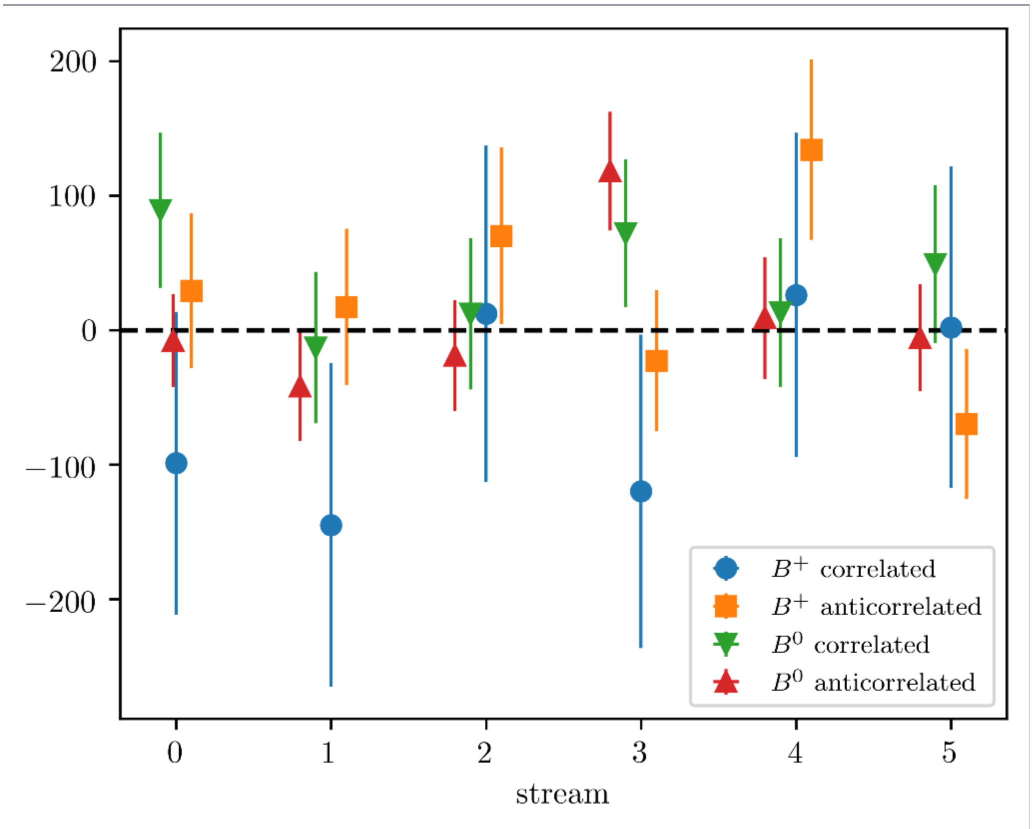


Figure (3.39) Residuals of the fitted signal yields resulting from the simultaneous 2D fit by stream and decay mode.

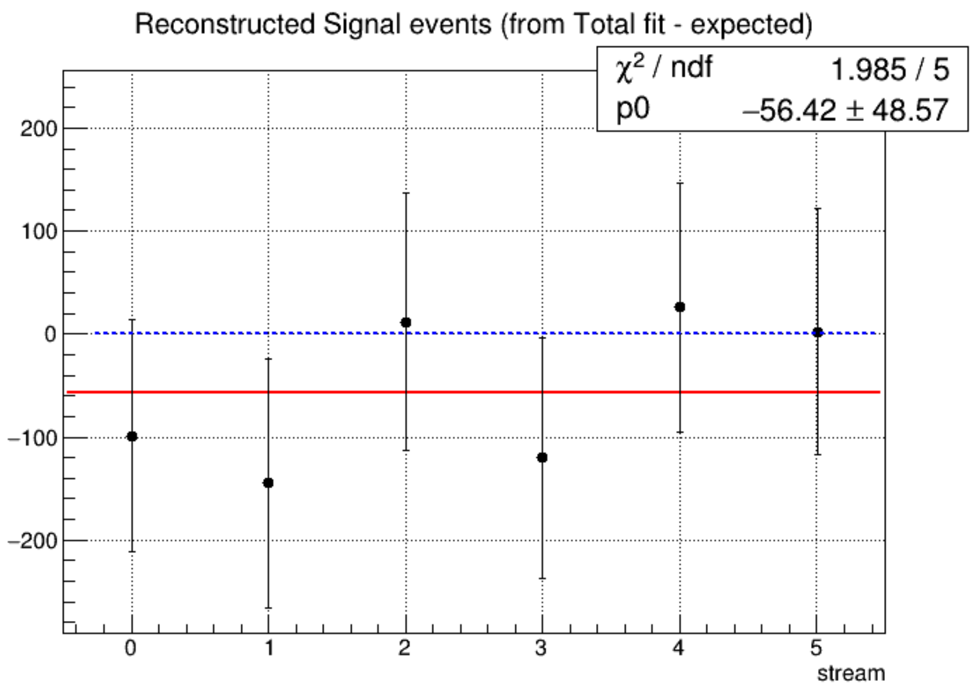


Figure (3.40) Residuals of the fitted signal yields of charged correlated decays.

3.2.6 toy Monte Carlo study

For the fit model also toy MC pseudo-experiments were performed in order to confirm the behavior of the fit setup. With toy MC experiments the yields, errors and the pulls of the fit are studied by generating our own pseudo-datasets, according to the MC (see plots in $3{\times}10^3$ pseudo-datasets are constructed, where each dataset was generated with the expected amount of events, distributed according to the Poisson distribution. Then the composition of each toy pseudo-experiment is fitted as if they were data, and the pull-value distributions of the fit results are calculated. From the plots showing the pull distributions of the fitted signal yields one can conclude that no bias is present.

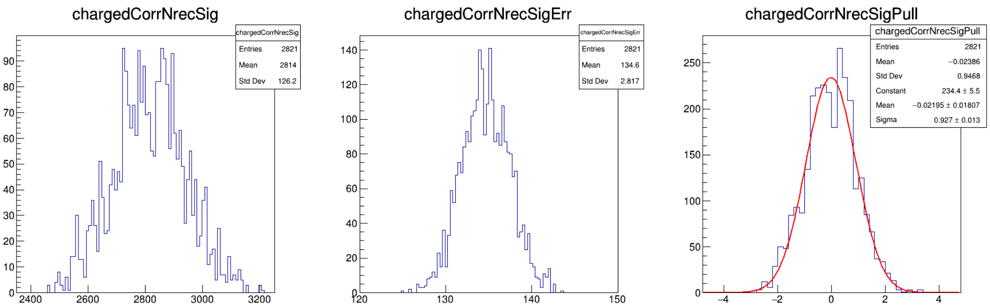


Figure (3.41) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the charged correlated decays.

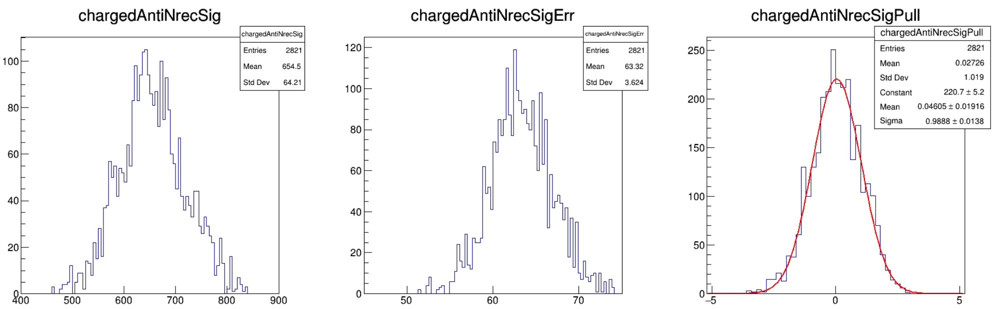


Figure (3.42) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the charged anticorrelated decays.

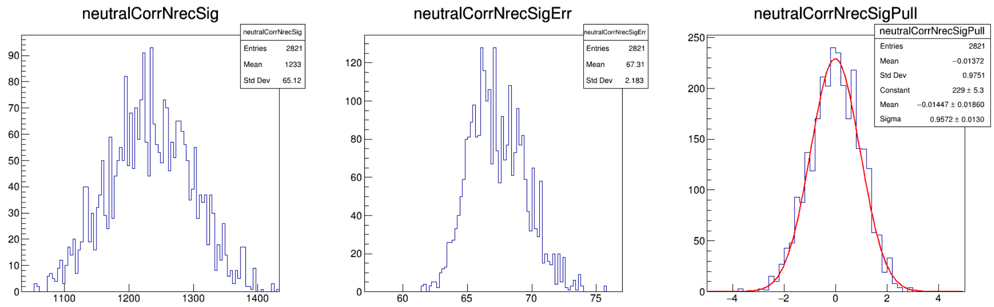


Figure (3.43) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the neutral correlated decays.

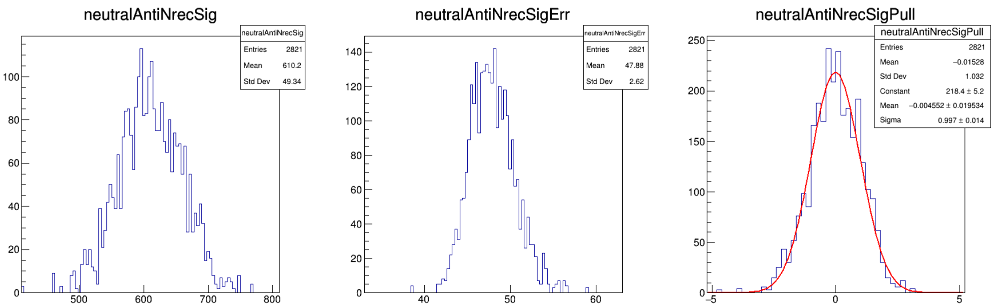


Figure (3.44) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits for the neutral anticorrelated decays.

Chapter 4

B_{tag} fit

The normalization for the branching fractions is determined from the number of correctly tagged B mesons. This value is again determined by fitting the M_{bc} distribution of all B meson candidates tagged by FEI regardless of the signal side B meson decay.
The tagged B meson candidates are selected according to the selection criteria used for the M_{bc} variable already illustrated in subsections Secs. 2.4.1 to 2.4.4.

4.1 Probability Density Functions (PDFs) for the B_{tag}

The reconstructed events can be categorized as follows:

- correctly reconstructed B meson candidates: **reconstructed signal**
- misreconstructed B meson candidates: **misreconstructed signal**
- B^0 mesons misreconstructed as B^+ (and vice-versa): **crossfed background**
- and **continuum background**

4.1.1 Total Signal fits

As in the 2D fits the total signal is distinguished in **reconstructed** and **misreconstructed signal** depending on wether it's peaking or not in M_{bc}.
Fig. 4.1 shows the fit on tagged charged B mesons corresponding to the normalization for the branching fractions of $B^- \to \Lambda_c^+$ decays. The M_{bc} distribution of the tagged B mesons is fitted with a combination of Crystal Ball and Gaussian as for the "peaking" component and the "flat" component is fitted with a Novosibirsk function. For the other decay channels instead an Argus function is used to describe the misreconstructed signal (see Figs. 6.4 to 4.4).

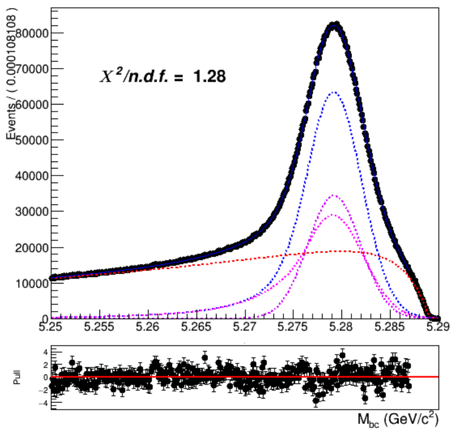


Figure (4.1) Fitted distribution of tagged charged B mesons in the charged correlated decays sample: reconstructed signal events are described by the blue dotted PDF, the misreconstructed with a Novosibirsk function (red dotted).

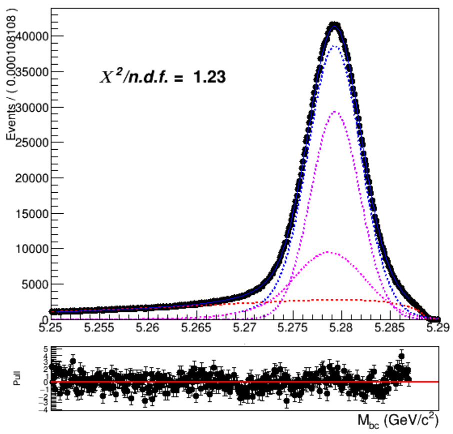


Figure (4.2) Fitted distribution of tagged charged B mesons in the charged anticorrelated decays sample.

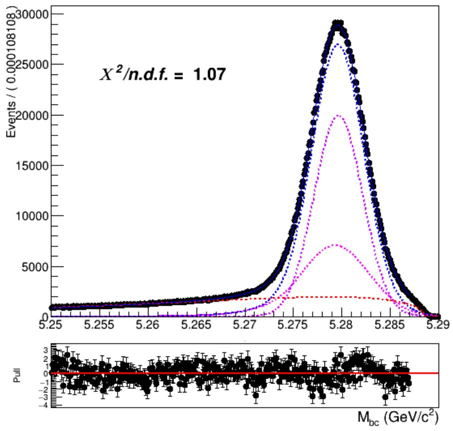


Figure (4.3) Fitted distribution of tagged neutral B mesons reconstructed in neutral correlated decays.

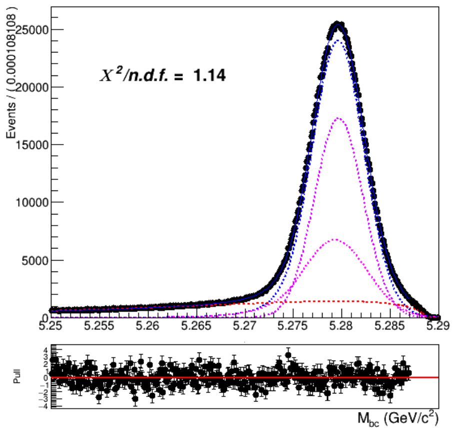


Figure (4.4) Fitted distribution of tagged charged B mesons reconstructed in neutral anticorrelated decays.

4.1.2 Crossfeed PDF

The crossfeed background is always fitted instead with a sum of a Novosibirsk and an asymmetric Gaussian PDF.

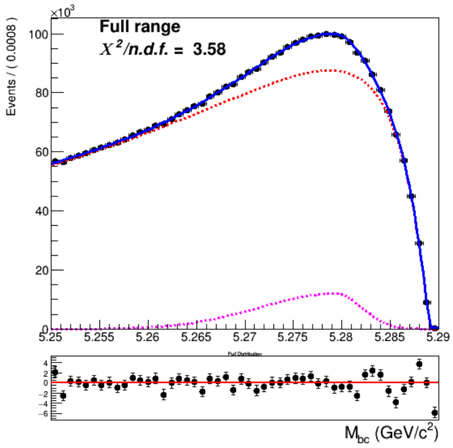


Figure (4.5) Crossfeed distribution of charged correlated decays fitted with a sum of Novosibirsk (red) and asymmetric Gaussian PDF (magenta)

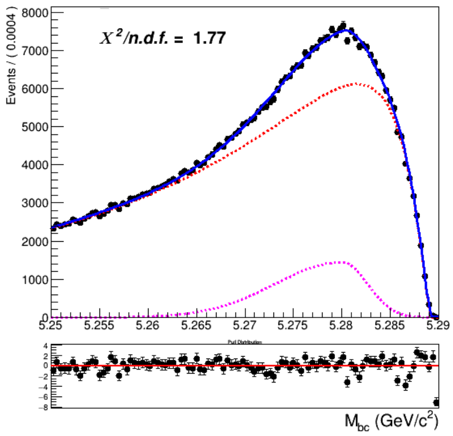


Figure (4.6) Crossfeed distribution of charged anticorrelated decays

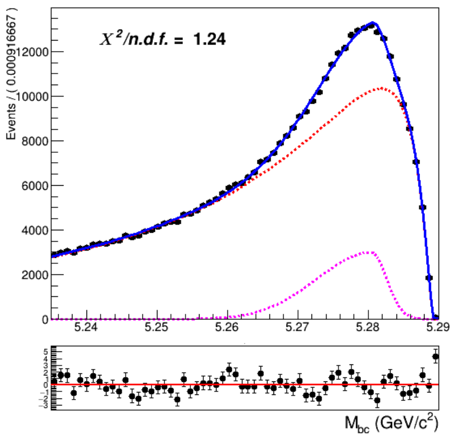


Figure (4.7) Crossfeed distribution of neutral correlated decays

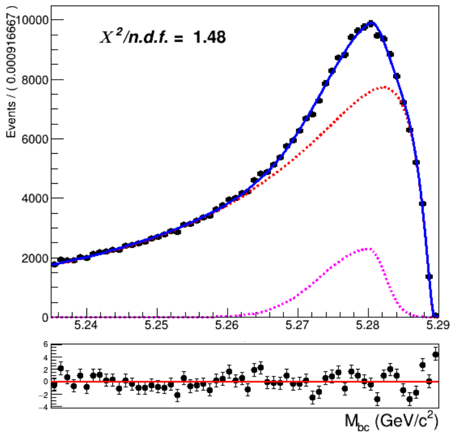


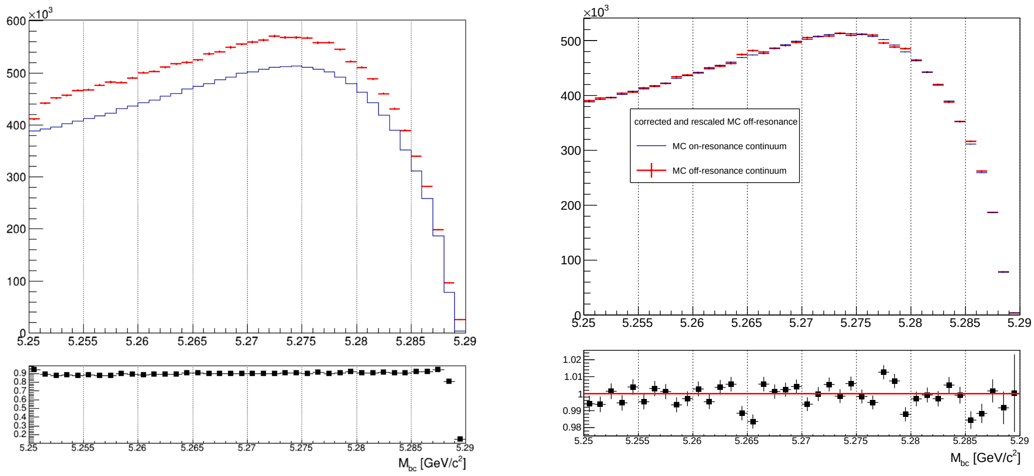
Figure (4.8) Crossfeed distribution of neutral anticorrelated decays

4.1.3 Continuum PDF

As for the continuum background, a similar procedure as the one described already for the two dimensional fit was adopted:

- first the off-resonance sample is scaled accordingly
- the ratio between the scaled off-resonance and the on-resonance in MC is calculated in each bin (see Fig.4.9a)
- the bin-correction is applied on an independent stream and the scaled and bin-corrected M_{bc} distribution is compared with the on-resonance distribution as shown in Fig.4.9b

Being the statistics much larger than in the 2D sample, there's no need to remove the continuum suppression cut on the off-resonance sample.



(a) (b)

Figure (4.9) On the left: M_{bc} distributions of the MC off-resonance sample and the MC continuum sample with applied continuum suppression. On the right: M_{bc} distributions of the corrected scaled MC off-resonance and on-resonance MC continuum.

4.2 B_{tag} fit

An independent Monte Carlo stream was used to test the total fit model on tagged B meson candidates. As in the 2D fit, the parameter for the width, σ_{CB}, of the Crystal Ball is floated. The ratio between expected crossfeed background events and misreconstructed signal events is fixed from the MC. The misreconstructed signal PDF is also not fully constrained: the parameter describing the tail is free. To avoid introducing significant systematic uncertainties in the fit deriving from the M_{bc} endpoint region, where one has a smearing effect due to variations of the beam energy at the MeV level, the range for the fit is restricted to values betweeen 5.250 and 5.287 GeV/c^2.

4.2.1 Fit results for $B^- \to \Lambda_c^+$ decays

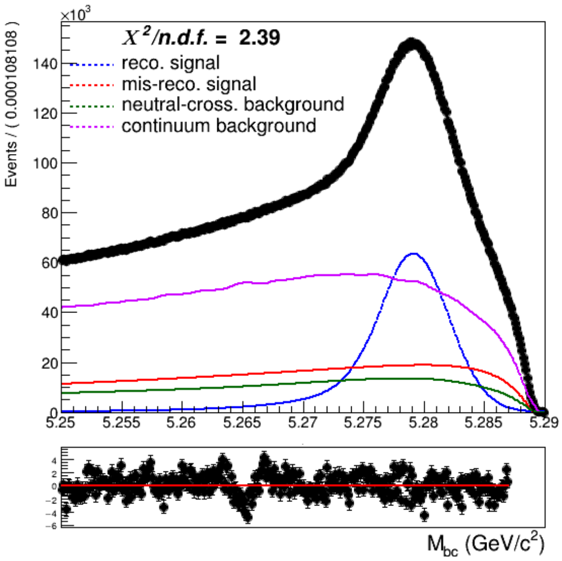


Figure (4.10) Total fit of tagged B mesons on Monte Carlo simulated data.

Yields for the reconstructed and misreconstructed signal are obtained from the fit:

	Fit results	expected
NrecSig	$4.7787 \cdot 10^{6} \pm 6748$	$4.7571 \cdot 10^{6} \pm 3214$
NmisSig	$5.3987 \cdot 10^{6} \pm 5617$	$5.4035 \cdot 10^{6} \pm 3376$

The normalization for the branching fraction is given by the NrecSig yields. The yields obtained in the fit differ by $\sim 3.2\sigma$ from the expected ones (obtained from a fit to the Total Signal events only). This discrepancy will impact the final measurement.

To check the stability of the fit model a toy MC study was performed with 3×10^3 pseudo-datasets. No evidence for possible biases in the reconstructed signal yields was found (see Fig. 4.11).

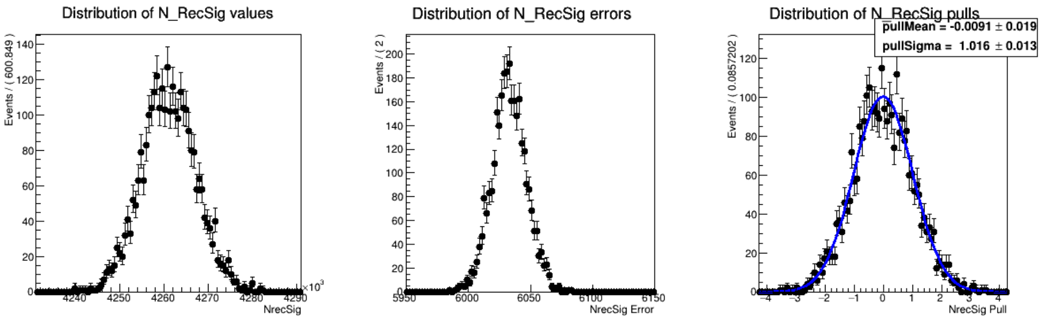


Figure (4.11) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits. (see Appendix **??** for the other free parameters' results)

4.2.2 Fit results for $B^- \to \bar{\Lambda}_c^-$ decays

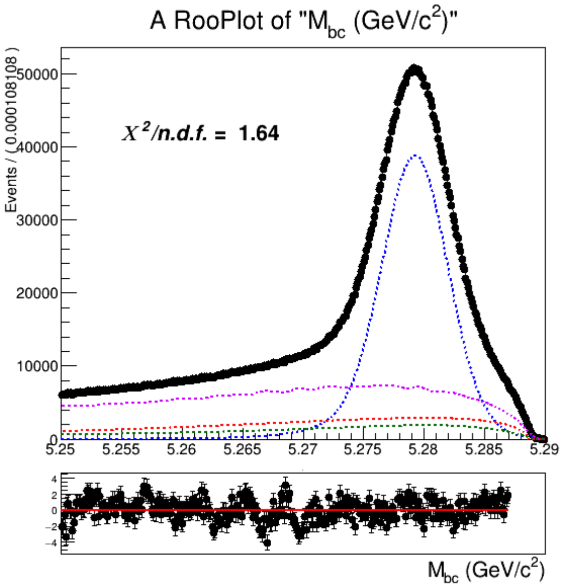


Figure (4.12) Total fit of tagged B mesons on Monte Carlo simulated data.

NrecSig	$2.50831{\cdot}10^6 \pm 11866$
NmisSig	$7.37010{\cdot}10^5 \pm 7472$

The Total Signal (the sum NrecSig+NmisSig) is 3292168 $\pm$ 2423 (to be compared with 3299629 from the Monte Carlo), which means a $\sim 3\sigma$ underestimation. As in the case of charged flavor-correlated decays, this can produce some systematic effect which needs to be taken into account.

But when performing a toy Monte Carlo study[1] the result show no bias on the recosntructed signal yields (see Fig. 4.13)

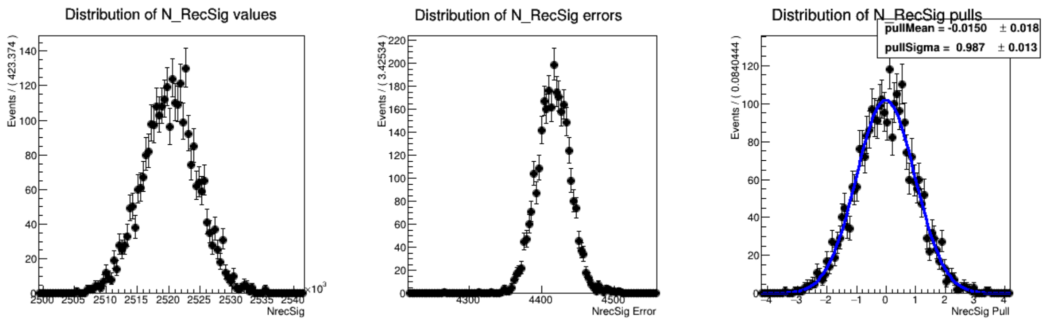


Figure (4.13) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits.

4.2.3 Fit results for $\bar{B}^0 \to \Lambda_c^+$ decays

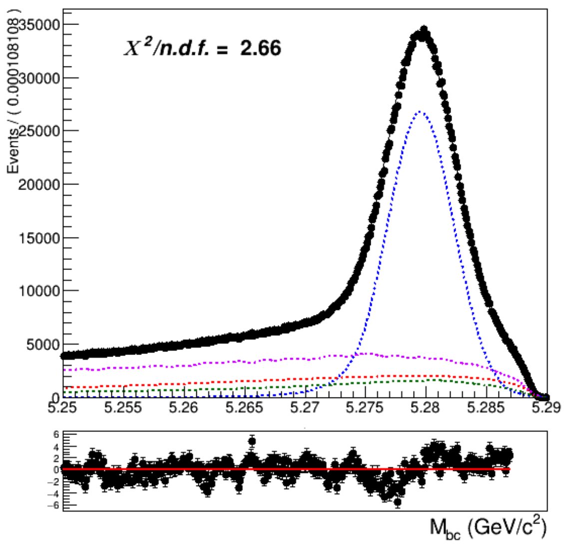


Figure (4.14) Total fit of tagged B mesons on Monte Carlo simulated data.

[1]as usual performed with 3×10^3 pseudo-datasets

Reconstructed and misreconstructed signal yields obtained from the fit:

NrecSig	$1.7215 \cdot 10^{6} \pm 3421$
NmisSig	$5.5950 \cdot 10^{5} \pm 2215$

The Total Signal (the sum NrecSig+NmisSig) is 2281033 $\pm$ 1947 (to be compared with 2286964 from the Monte Carlo). Also in this case there's a $\sim 3\sigma$ underestimation.

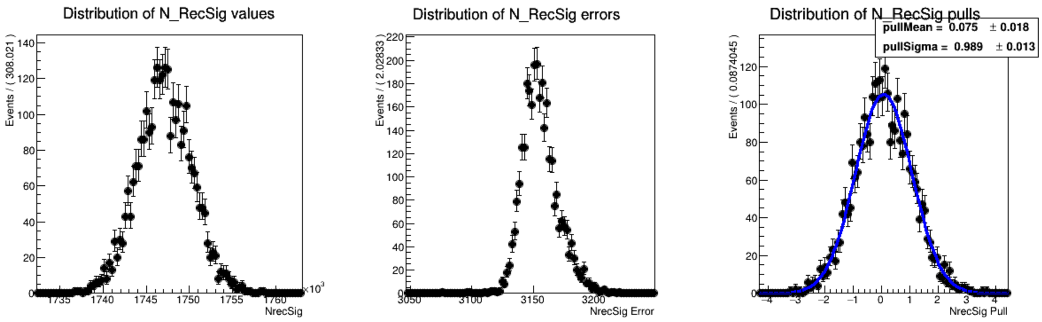


Figure (4.15) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits.

4.2.4 Fit results for $B^0 \to \bar{\Lambda}_c^+$ decays

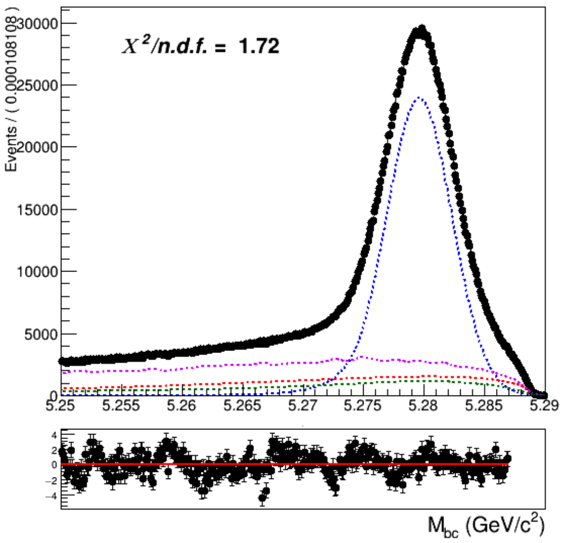


Figure (4.16) Total fit of tagged B mesons on Monte Carlo simulated data.

Reconstructed and misreconstructed signal yields obtained from the fit:

NrecSig	$1.5302 \cdot 10^{6} \pm 3269$
NmisSig	$3.8332 \cdot 10^{5} \pm 2072$

The Total Signal (the sum NrecSig+NmisSig) is 1913476 ± 1812 (to be compared with 1920156 from the Monte Carlo). Also in this case there is an underestimation of about a $\sim 3.7\sigma$.

The toy MC study result for the signal yields is shown in Fig. 4.17. Also in this case no hints of bias are visible.

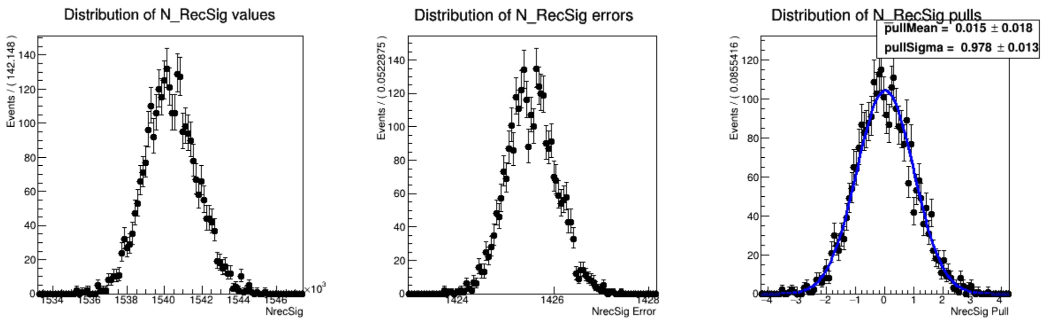


Figure (4.17) Plots showing distributions of the fitted signal yields, errors and the pull distribution of all pseudo-fits.

Chapter 5

Efficiencies

Here a decision was made not to rely on the estimated number of B meson pair, as it is usually done, and the absolute FEI efficiency, since the latter shows large discrepancy between MC and data (see i.e. the results reported in the PhD Thesis by M. Gelb [4] and also by J. Schwab [5]) and also it depends strongly on the signal-side (i.e. $\epsilon^{+}_{FEI} \neq \epsilon^{+}_{FEI,sig}$). Instead, to limit the systematics, the branching ratio normalization is obtained using the fitted tagged B mesons and the ratio $\epsilon^{+}_{FEI,sig}/\epsilon^{+}_{FEI}$ measured on MC, which one can expect to be better described by the MC than the absolute FEI efficiency.

5.1 FEI Efficiencies

One needs to distinguish two different FEI efficiencies as already anticipated.

The generic FEI efficiency $\epsilon_{FEI} = \frac{n^{\circ} of tagged B}{total n^{\circ} of events}$ is the generic efficiency to tag correctly a B^+B^- or $B^0\bar{B}^0$ event, i.e.: the hadronic decay-chain of the tag-side B meson is correctly reconstructed, independently from signal-side. The denominator is given by all $B\bar{B}$ events present in the considered sample. The generic FEI efficiency is the one that affects the normalization in the branching ratio calculation.

The FEI efficiency that affects the numerator of the branching ratio is called signalside-dependent efficiency $\epsilon_{FEI,sig}$. In the case one is trying to reconstruct $B^- \to \Lambda_c^+$ decays, then this efficiency is determined as the follwoing ratio: $\epsilon^{+}_{FEI,sig} = \frac{n^{\circ} of tagged B^{+/-} of B^- \to \Lambda_c^+}{total n^{\circ} of B^- \to \Lambda_c^+ events}$

5.1.1 Generic FEI efficiency

The number of correctly tagged B mesons is determined from the yields of reconstructed signal events obtained from a fit of the B_{tag} distributions shown already in Sec. 4.1.1. As an example Fig. 5.1 shows the same distributions shown in Fig. 4.1 where all shaping parameters have been fixed and only the the normalisations of **reconstructed** and

misreconstructed signal were floated.

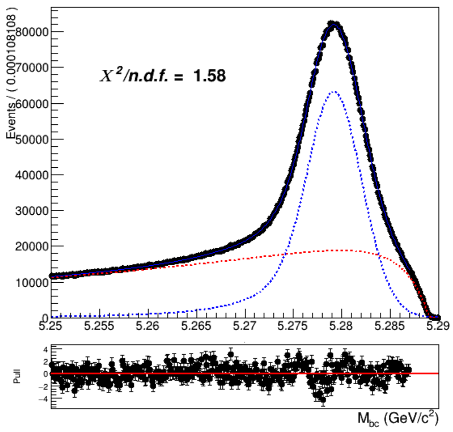


Figure (5.1) Fitted distribution of tagged charged B mesons in the charged correlated decays sample.

5.1.2 Signalside-dependent FEI efficiency

The FEI tag-side efficiency for signal events is calculated upon 10 streams of on-resonance Monte Carlo streams of simulated data in order to minimize statistical uncertainties. As an example Fig. 5.2 shows the fit performed on the tagged B mesons having in the ROE a companion decaying $B^- \to \Lambda_c^+$ (charged correlated decays). Dividing the yields obtained by the fit by the total number of $B^- \to \Lambda_c^+ events$ present in the 10 streams it is possible to determine the FEI tag-side efficiency for signal events.

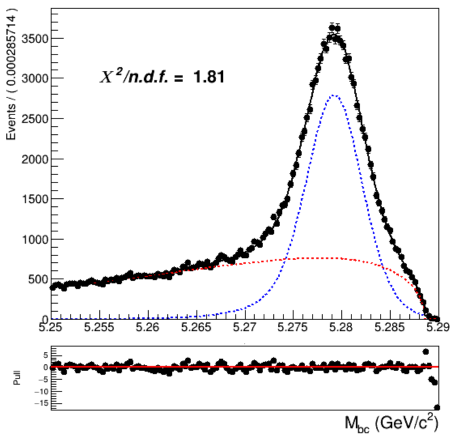


Figure (5.2) Fit of tagged B mesons in the "signal events" sample

5.2 Λ_c efficiency

The efficiency of reconstructing the Λ_c baryon after correctly tagging the charged B meson, can be estimated from Monte Carlo simulated data. All available 10 streams of on-resonance Monte Carlo streams of simulated data were used to minimize statistical uncertainties. The efficiency is evaluated as follows from the fraction:

$$\frac{N_{recSig}(B_{tag}, \Lambda_c)}{N_{recSig}(B_{tag}^{sig})} \tag{5.1}$$

where $N_{recSig}(B_{tag}, \Lambda_c))$ are the yields of reconstructed signal from a two dimensional fit of corresponding inclusive decay. and $N_{recSig}(B_{tag}^{sig})$ are the yields of correctly reconstructed signal from the same fit of tagged B mesons used to calculate the Signalside-dependent FEI efficiency.

5.2.1 $B^- \to \Lambda_c^+$ decays

$\epsilon^+_{FEI,sig} = 0.636 \pm 0.003\%$ (calculated with 10 streams)
$\epsilon^+_{FEI} = 0.6379 \pm 0.0004\%$ $\frac{\epsilon^+_{FEI,sig}}{\epsilon^+_{FEI}} = 0.997 \pm 0.005$
$\epsilon_{\Lambda_c} = 37.78 \pm 0.45$ %

5.2.2 $B^- \to \bar{\Lambda}_c^-$ decays

$\epsilon^+_{FEI,sig} = 0.334 \pm 0.003\%$ (calculated with 10 streams)
$\epsilon^+_{FEI} = 0.3461 \pm 0.0002\%$ FEI calculation includes mixed events, ϵ_{Λ_c} is calculated without them $\frac{\epsilon^+_{FEI,sig}}{\epsilon^+_{FEI}} = 0.965 \pm 0.009$
$\epsilon_{\Lambda_c} = 42.39 \pm 0.66$ %

5.2.3 $\bar{B}^0 \to \Lambda_c^+$ decays

$\epsilon^0_{FEI,sig} = 0.247 \pm 0.002\%$ (calculated with 10 streams unmixed)
$\epsilon^0_{FEI} = 0.234 \pm 0.0002\%$ (1 unmixed stream)

using only unmixed:
$\frac{\epsilon^0_{FEI,sig}}{\epsilon^0_{FEI}} = 1.056 \pm 0.009$
$\epsilon_{\Lambda_c} = 39.59 \pm 0.49$ % (using unmixed events)

5.2.4 $\bar{B}^0 \to \Lambda_c^-$ decays

$\epsilon^0_{FEI,sig} = 0.210 \pm 0.003\%$ (calculated with 10 streams unmixed)
$\epsilon^0_{FEI} = 0.206 \pm 0.0002\%$ (1 unmixed stream)

using only unmixed:

$$\frac{\epsilon^0_{FEI,sig}}{\epsilon^0_{FEI}} = 1.017 \pm 0.015$$

$$\frac{\epsilon^+_{FEI,sig}}{\epsilon^+_{FEI}} = 1.005 \pm 0.014$$

$$\epsilon_{\Lambda_c} = 41.49 \pm 1.13\ \%$$

Chapter 6

Branching ratio

To measure the inclusive branching fractions

$$Br(B^{+/-} \to \Lambda_c^+ X) = \frac{N_{tag,\Lambda_c} \cdot \epsilon^+_{FEI}}{N_{tag} \cdot Br(\Lambda_c^+ \to pK^-\pi^+)\epsilon_{\Lambda_c}\epsilon^+_{FEI,sig}} \tag{6.1}$$

$$Br(B^0 \to \Lambda_c^{+/-} X) = \frac{N_{tag,\Lambda_c} \cdot \epsilon^0_{FEI}}{N_{tag} \cdot Br(\Lambda_c^+ \to pK^-\pi^+)\epsilon_{\Lambda_c}\epsilon^0_{FEI,sig}} \tag{6.2}$$

the following quantities need to be known:

- N_{tag,Λ_c} is the reconstructed signal yield obtained from a two dimensional fit of M_{bc} and $M(pK\pi)$ in the final sample.
- N_{tag} is the reconstructed signal yield obtained from the M_{bc} fit of all the tagged B mesons in the final sample.
- ϵ_{Λ_c} is the Λ_c reconstruction efficiency.
- ϵ^+_{FEI} (ϵ^0_{FEI}) represent the hadronic tag-side efficiency for generic B^+B^- ($B^0\bar{B}^0$) events.
- $\epsilon^+_{FEI,sig}$ ($\epsilon^0_{FEI,sig}$) represents the hadronic tag-side efficiency for B^+B^- ($B^0\bar{B}^0$) events where the tagged B meson decays hadronically and the accompanying meson decays inclusively into the studied signal channel.
- $Br(\Lambda_c^+ \to pK^-\pi^+)$: the branching fraction of the decay mode used to reconstruct the Λ_c baryon.

	total fit	signal fit	BELLE MC VALUE
stream 0	$(2.84 \pm 0.13)\%$	$(2.96 \pm 0.07)\%$	$(2.91 \pm 0.03)\%$
stream 1	$(2.82 \pm 0.14)\%$	$(2.99 \pm 0.07)\%$	$(2.91 \pm 0.03)\%$
stream 2	$(2.97 \pm 0.14)\%$	$(2.97 \pm 0.07)\%$	$(2.90 \pm 0.03)\%$
stream 3	$(2.76 \pm 0.14)\%$	$(2.90 \pm 0.07)\%$	$(2.91 \pm 0.03)\%$
stream 4	$(3.06 \pm 0.14)\%$	$(3.05 \pm 0.07)\%$	$(2.90 \pm 0.03)\%$
stream 5	$(2.83 \pm 0.14)\%$	$(2.84 \pm 0.07)\%$	$(2.92 \pm 0.03)\%$
average	$(2.88 \pm 0.06)\%$	$(2.95 \pm 0.03)\%$	$(2.91 \pm 0.01)\%$

Table (6.1) charged corr

	total fit	signal fit	BELLE MC VALUE
stream 0	$(1.24 \pm 0.11)\%$	$(1.19 \pm 0.05)\%$	$(1.217 \pm 0.002)\%$
stream 1	$(1.24 \pm 0.11)\%$	$(1.21 \pm 0.05)\%$	$(1.218 \pm 0.002)\%$
stream 2	$(1.35 \pm 0.12)\%$	$(1.23 \pm 0.05)\%$	$(1.218 \pm 0.002)\%$
stream 3	$(1.16 \pm 0.10)\%$	$(0.20 \pm 0.05)\%$	$(1.215 \pm 0.002)\%$
stream 4	$(1.44 \pm 0.13)\%$	$(1.21 \pm 0.05)\%$	$(1.218 \pm 0.002)\%$
stream 5	$(1.04 \pm 0.11)\%$	$(1.15 \pm 0.05)\%$	$(1.217 \pm 0.002)\%$
average	$(1.25 \pm 0.05)\%$	$(1.20 \pm 0.02)\%$	$(1.217 \pm 0.001)\%$

Table (6.2) charged anticorr

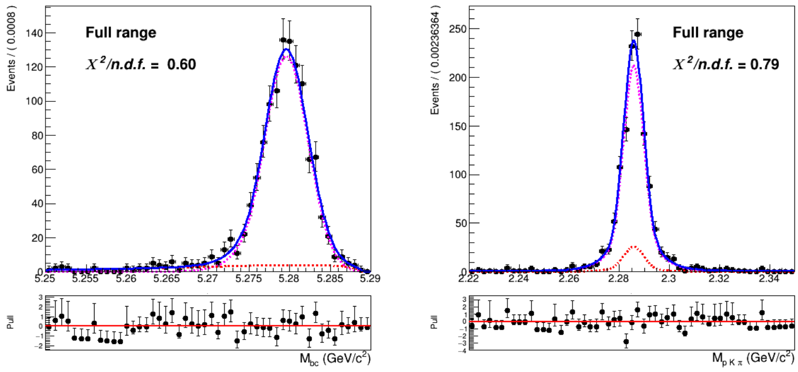


Figure (6.1) Unmixed neutral correlated total signal events in M_{bc} and $M(pK\pi)$

6.1 extraction of neutral decays branching fractions

Due to the mixing phenomenon of neutral B mesons the measurement requires an intermediate step in order to correct the production obtained rates of Λ_c baryons in the reocoil of the reconstructed neutral B mesons for the $B^0\bar{B}^0$ mixing.
Being χ_d the mixing parameters, on average, if one looks at the recoil of a correctly reconstructed B^0 $(\bar{B}^0)$ meson $1-\chi_d$ of the time the reconstructed Λ_c baryon will have decayed from a $\bar{B}^0$ (B^0) and χ_d of the time from B^0 $(\bar{B}^0)$, due to mixing.

i.e.:

$$\begin{cases} \mathcal{B}(B^0_{tag}, \Lambda_c^-) = (1-\chi_d)\mathcal{B}_c + \chi_d\bar{\mathcal{B}}_c \\ \mathcal{B}(\bar{B}^0_{tag}, \Lambda_c^-) = (1-\chi_d)\bar{\mathcal{B}}_c + \chi_d\mathcal{B}_c \end{cases} \tag{6.3}$$

are the two equations corresponding to the spurious correlated/ anticorrelated branching ratios, with χ_d factor of missing events and χ_d amount of events that belong actually to the other type of decays. Solving those simultaneous equations one obtains the mixing-corrected branching ratios $\mathcal{B}_c$ (correlated decays) and $\bar{\mathcal{B}}$ (anticorrelated decays):

$$\begin{cases} \mathcal{B}_c = \dfrac{\mathcal{B}(B^0_{tag}, \Lambda_c^-) - \chi_d(\mathcal{B}(B^0_{tag}, \Lambda_c^-) + \mathcal{B}(\bar{B}^0_{tag}, \Lambda_c^-))}{1-2\chi_d} \\ \bar{\mathcal{B}}_c = \dfrac{(\bar{B}^0_{tag}, \Lambda_c^-) - \chi_d(\bar{B}^0_{tag}, \Lambda_c^-) + \mathcal{B}(B^0_{tag}, \Lambda_c^-))}{1-2\chi_d} \end{cases} \tag{6.4}$$

First a check of the decays that didn't experience mixing is performed. For this check one stream samples of signal events that didn't experience mixing were used to obtain signal yields (see Figures below).

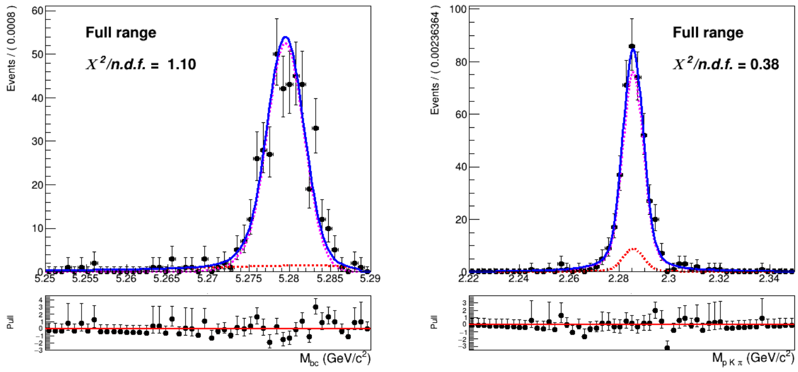


Figure (6.2) Unmixed neutral anticorrelated total signal events in M_{bc} and $M(pK\pi)$

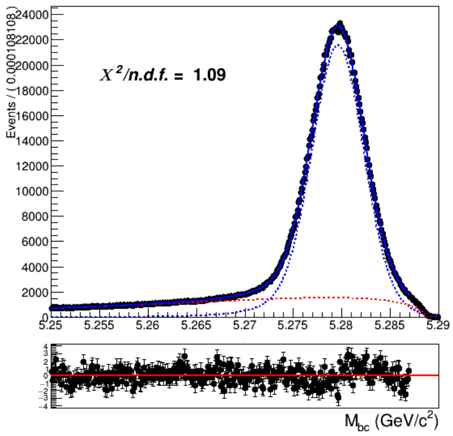


Figure (6.3) Unmixed neutral B mesons in the correlated decay sample.

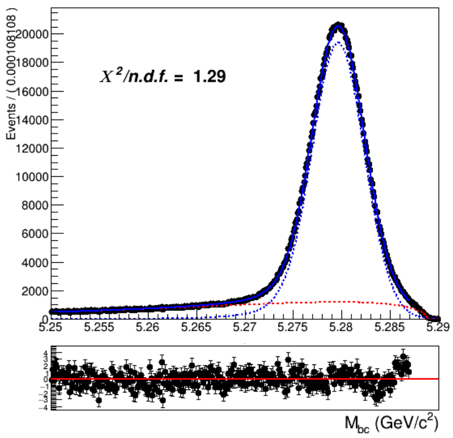


Figure (6.4) Unmixed neutral B mesons in the anticorrelated decay sample.

6.1.1 unmixed neutral decays branching ratios

The following two tables report the values of branching fractions for the neutral decays' events that didn't experience mixing (i.e.: $\mathcal{B}_c$ and $\bar{\mathcal{B}}_c$), obtained with the fits shown in

	signal fit	BELLE MC VALUE
stream 0	$(3.48 \pm 0.11)\%$	$(3.439 \pm 0.003)\%$
stream 1	$(3.54 \pm 0.11)\%$	$(3.431 \pm 0.003)\%$
stream 2	$(3.48 \pm 0.11)\%$	$(3.433 \pm 0.003)\%$
stream 3	$(3.47 \pm 0.11)\%$	$(3.433 \pm 0.003)\%$
stream 4	$(3.34 \pm 0.11)\%$	$(3.434 \pm 0.003)\%$
stream 5	$(3.45 \pm 0.11)\%$	$(3.436 \pm 0.003)\%$
average	$(3.46 \pm 0.05)\%$	$(3.434 \pm 0.001)\%$

Table (6.3) Measured branching fraction values obtained using only unmixed events for the six different streams (only statistical uncertainties are displayed) and its average.

	signal fit	BELLE MC VALUE
stream 0	$(1.35 \pm 0.09)\%$	$(1.300 \pm 0.002)\%$
stream 1	$(1.30 \pm 0.07)\%$	$(1.299 \pm 0.002)\%$
stream 2	$(1.31 \pm 0.07)\%$	$(1.300 \pm 0.002)\%$
stream 3	$(1.31 \pm 0.07)\%$	$(1.299 \pm 0.002)\%$
stream 4	$(1.22 \pm 0.09)\%$	$(1.295 \pm 0.002)\%$
stream 5	$(1.30 \pm 0.07)\%$	$(1.298 \pm 0.002)\%$
average	$(1.30 \pm 0.03)\%$	$(1.299 \pm 0.001)\%$

Table (6.4) Measured branching fraction values obtained using only unmixed events for the six different streams (only statistical uncertainties are displayed) and its average.

The good agreement within statistical errors of the obtained branching fractions excluding mixing events guarantees that the procedure to obtain the branching fractions is correct and any possible discrepancy observed when mixing is included does not depend on it.

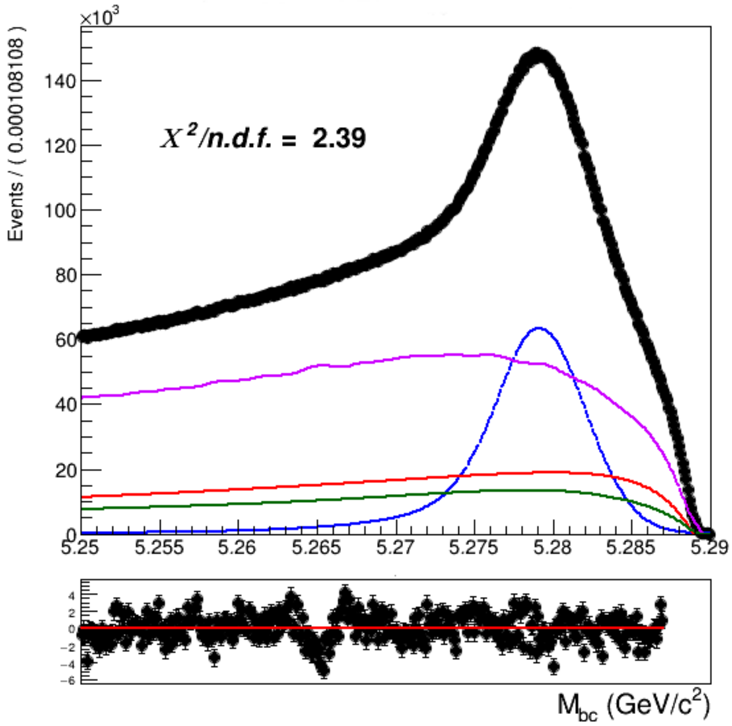
6.2 Systematics

6.2.1 Continuum background

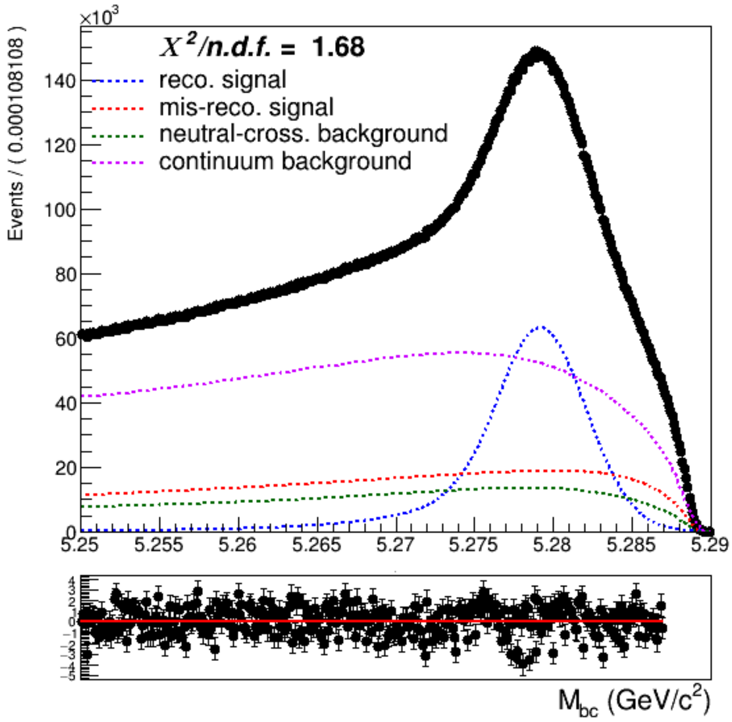
2 sources of systematics:

- difference in the shapes
- different impact of the Continuum Suppression cut

To estimate the first type of uncertainty two-dimensional fits with varied parameters' values by their uncertainties (a fit with +err and −err) were performed.
The rescaling procedure used to model the on-resonance continuum distribution is far from perfect, especially in the case of the B_{tag} fit. As one can see from Fig. 6.5a:



(a)



(b)

Figure (6.5) Above: Fitted M_{bc} of charged correlated decays with scaled off-resonance continuum distribution. Below: same M_{bc} distribution with the true continuum distribution.

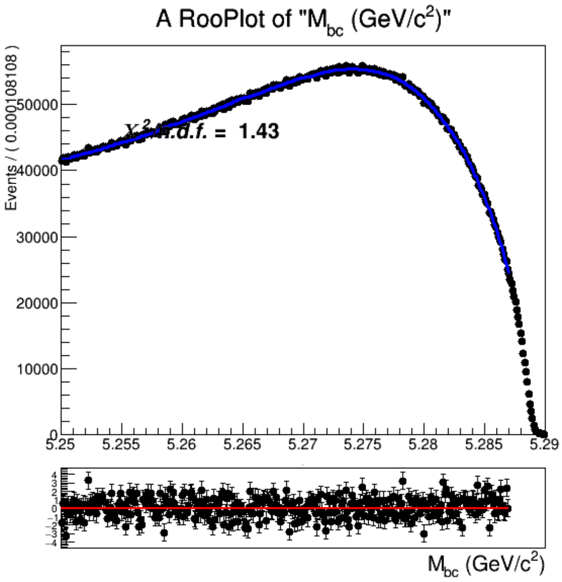


Figure (6.6) M_{bc} on-resonance MC distribution of charged correlated decays.

The bumps visible for $M_{bc} < 5.27 GeV/c^2$ in Fig. 6.5a are a consequence of the imperfect scaling procedure used to model the continuum background. In fact, they disappear completely (in Fig. 6.5b) when the same fit is performed using the pdf obtained directly from the on-resonance distribution (Fig. 6.6) There is a significant impact on the fit results, the values from the B_{tag} fit in the charged correlated channel are reported here below:

	Fit with scaled continuum	Fit with true continuum
N_{recSig} yields	$4.7787 \cdot 10^6 \pm 6.75 \cdot 10^3$	$4.7474 \cdot 10^6 \pm 8.01 \cdot 10^3$

Table (6.5) Signal yields obtained in B_{tag} fits shown in Fig. 6.5a and Fig. 6.5b.

The yields are to be compared with the values obtained when fitting the total signal distribution shown in Fig. 4.1: $N_{recSig} = 4.7571 \cdot 10^6 \pm 3.23 \cdot 10^3$. The continuum modeling alone is responsible for a discrepancy of about 3.2σ on the yields. This discrepancy and the impact on the branching fraction are accounted in the systematic uncertainties derived from the continuum modeling. This source of systematics is responsible for 0.02 % uncertainty on the branching ratio. The other type of systematic uncertainty in modeling the continuum is originated by the continuum suppression cut having a slightly different efficiency on data (as a consequence of the shift in off-resonance MC and data visible in the the $foxWolframR2$ distribution in Fig. 3.19). As already discussed, it originates a possible discrepancy of about 2.25% in continuum background events in the two dimensional fit (and only 1.25% in the B_{tag} fit). The statistical uncertainty on this fraction of events can also be taken into account as systematics. Being the number of events in the off-resonance data sample without the continuum suppression applied is very small, the uncertainty in the mentioned fraction of events is negligible compared to the statistical uncertainty on the on-resonance continuum background events in MC: it would account for 0.002% on the BR value. Therefore, this second source of uncertainty is not taken into account. There would be also a third of systematic uncertainty given by potential difference in the on-/off-resonance correction between data and MC, but there's no way one can estimate it properly.

6.2.2 Crossfeed ratio

The ratio crossfeed/misreconstructed signal is kept fixed in the B_{tag} fit to the MC value. This choice was made according to the fact that the two categories of events have a similar origin: in both cases the B mesons were not correctly reconstructed, either because of missing or wrongly added particles (misreconstructed signal events) or, in the case of crossfeed events because the tagged B meson was not the required one (B^0 meson instead of a $B^{+/-}$ meson). The ratio between these two categories of events is therefore expected to be very similar in MC and data, though there's no guarantee that the efficiency to reconstruct them is the same in data in data and consequently the above mentioned ratios could differ on data. Unfortunately there's no direct way to have an estimate of the possible discrepancy for them.
In [4] (and previously in [5]) it was found that there's a substantial difference in terms of tagging efficiency for FEI applied on Monte Carlo and on Belle data, being the discrepancy around $\sim 20\%$. We can assume that the efficiency for the two categories of events on data will both differ of that value and the ratio of the events being the same MC value, but in absence of any other method to estimate the uncertainty on it one can consider a maximal discrepancy of 20% between Monte Carlo and data to study the impact on the yields[1].

6.2.3 Crossfeed B_{tag} PDF

Since also the shapes of the PDFs decribing the crossfeed background in the B_{tag} fit are fully fixed to the ones determined with the limited Monte Carlo statistics, also their statistical uncertainties need to be taken into account as possible source of systematics. To estimate this source of uncertainty fits with varied parameters' values by their uncertainties (a fit with $+$err and $-$err) were performed.

6.2.4 2DFit crossfeed normalization

As discussed in the 2D simulateous fit section, in each decay channel the normalization of crossfeed events is estimated using the fitted reconstructed signal events of the

[1] This method was also validated with the control decay sample and the originated uncertainty is well within the PDG reported ones.

corresponding crossfeeding channel in the as follows:

$$N_{cross} = N_{recSig}^{correlated}(\frac{\epsilon_{cross}}{\epsilon_{recSig}})^{correlated} + N_{recSig}^{anticorrelated}(\frac{\epsilon_{cross}}{\epsilon_{recSig}})^{anticorrelated} \tag{6.5}$$

where ϵ_{recSig} and ϵ_{cross} are respectively the efficiency to reconstruct correctly signal events in the crossfeeding channel and the efficiency of reconstructing those events as crossfeed in the channel of interest.
The uncertainties of those two efficiencies may cause an overestimation or underestimation of the crossfeed in the channel of interest. To estimate the systematic uncertainty deriving from it, the simultaneous fits have been performed modifying the ratio of the efficiencies by the respective uncertainties maximising the shift, i.e. using $\frac{\epsilon_{cross}+\sigma}{\epsilon_{recSig}-\sigma}$ and $\frac{\epsilon_{cross}-\sigma}{\epsilon_{recSig}+\sigma}$

6.2.5 2DFit M_{bc} peaking/flat PDFs

As discussed in **??** the shapes of the PDFs describing the generic background (peaking or flat in M_{bc}) are kept fixed in the two dimensional fit. This source of systematics is handled in the same way as the crossfeed background in the B_{tag}: fits with varied parameters' values by their uncertainties were performed (procedure is repeated for M_{bc} peaking and flat background).

6.2.6 PID efficiency correction

The PID selection efficiency for the three charged particles in the signal decay needs to be corrected on MC due to various differences, when comparing to data. The Belle PID group has prepared a set of correction factors and tables of systematic uncertainties for PID efficiencies for all charged particles. The proton identification efficiency was studied in [6]. The inclusive Λ^0 decay $\Lambda^0 \to p\pi^-$ was used to examine the proton identification efficiency difference between data and MC in *Belle*. The datasets for the SVD1 and SVD2 periods are treated separately, and the efficiency ratio dependence on proton charge, momentum and polar angle is considered. The study is done for the proton ID cut values 0.6, 0.7, 0.8 and 0.9[2]. The binning on the momentum starts at 0.2 GeV. The proton ID efficiency is defined as

$\epsilon_{PID} = \frac{\text{number of} \, p \, \text{tracks identified as} \, p}{\text{number of} \, p \, \text{tracks}}$

and the comparison between MC efficiency and data efficiency by a double ratio defined as

$$R_p = \epsilon^{data}/\epsilon^{MC}$$

[2]Here, proton ID cut value X means $\mathcal{L}_{p/K} > X$ and $\mathcal{L}_{p/\pi} > X$

The average proton ID correction is estimated to be: $R_p = 0.969 \pm 0.003$.

The kaon identification efficiency was studied in detail in Belle Note 779 [7] (`http://belle.kek.jp/secured/belle_note/gn779/bn779.ps.gz`). The decay $D^{*+} \to D^0\pi^+$ followed by $D^0 \to K^-\pi^+$, was used to examine it. As for the proton identification efficiency it considers the dependence on Kaon charge, momentum and polar angle and same ID cut values[3]

6.2.7 $B^- \to \Lambda_c^+$ decays

Regarding the systematic uncertainties deriving from the continuum modeling in the two-dimensional fit are estimated as already described in **??** one obtains an absolute value of 0.04% on the branching fraction.
Instead the imperfect modeling of the continuum in the B_{tag} fit is responsible for a 0.02 % on the branching fraction in absolute value.
Continuum modeling uncertainties account for 0.05% total uncertainties on the branching fraction in absolute value.

To estimate the systematics derived from the crossfeed/misreconstructed signal ratio, in the B_{tag} fit the number of crossfeed events were varied artificially in order to have $\pm 20\%$ different ratio, keeping the previosuly determined Monte Carlo ratio fixed.

	$-\sigma$	$+\sigma$	$\pm\bar{\sigma}$
offset	20010	9330	14670

Table (6.6) Offsets on the N_{recSig} signal yields obtained varying of $\pm 20\%$ the ratio of crossfeed/misreconstructed events in the B_{tag} fit and mean deviations reported in the last column.

The offsets that a potential $\pm 20\%$ different crossfeed/misreconstructed signal ratio produce results in 0.01% systematic uncertainity on the branching fraction.
The systematic uncertainties deriving from the crossfeed background modeling in the B_{tag} fit is estimated following the same procedure described for continuum modeling systematics in the two dimensional fit.
In the table below the signal yields' offsets are listed changing the parameters within their uncertainties, and the mean offsets value used to calculate the expected uncertainty on the BR value. The resulting absolute systematic uncertainty is about 0.01% on the BR value.

[3]Here, Kaon ID cut value X means $\mathcal{L}_{K/\pi} > X$, so for values below X the tracks are identified as pions

	$-\sigma$	$+\sigma$	$\pm\bar{\sigma}$
offset	14810	17280	16045

Table (6.7) Offsets on the N_{recSig} signal yields obtained varying the parameters of crossfeed background PDFs within their uncertainties in the B_{tag} fit and mean deviations reported in the last column.

source	relative (%)	absolute (%)
Continuum modeling	1.77	0.05
B_{tag} fit Crossfeed fraction	0.35	0.01
B_{tag} fit Crossfeed PDF shape	0.35	0.01
2DFit crossfeed normalization	0.70	0.02
M_{bc} peaking PDF shape	0.35	0.01
M_{bc} flat PDF shape	0.35	0.01
$\epsilon^{+}_{FEI,sig}/\epsilon^{+}_{FEI}$	0.35	0.01
ϵ_{Λ_c}	1.41	0.04
PID	1.75	0.05
Tracking efficiency	0.35	0.01
Total	3.18	0.09

Table (6.8) Systematic uncertainties in the determination of the $B^+ \to \bar{\Lambda}_c^- X$ branching fraction in %.

6.2.8 $B^+ \to \Lambda_c^+$ decays

Regarding the systematic uncertainties deriving from the continuum modeling in the two-dimensional fit are estimated one obtains an absolute value of 0.02% on the branching fraction.

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